

FINITE- AND LARGE-SAMPLE INFERENCE FOR MODEL AND COEFFICIENTS IN HIGH-DIMENSIONAL LINEAR REGRESSION WITH REPRO SAMPLES

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Abstract In this paper, we present a novel and effective inference approach to conduct both finite- and large-sample inference for high-dimensional linear regression models. This approach is developed under the so-called *repro samples* framework, in which we conduct statistical inference by creating and studying the behavior of artificial samples that are obtained by mimicking the sampling mechanism of the data. We construct confidence sets for (a) the true model corresponding to the nonzero coefficients, (b) a single or any collection of regression coefficients, and (c) both the model and regression coefficients jointly. To facilitate the constructions of these confidence sets and overcome computational difficulties of searching all possible models, we use an innovative Fisher inversion technique to construct a model candidate set that includes the true sparse model with the probability close to 1 for models with both Gaussian and non-Gaussian errors. The proposed approach fills in two major gaps in the high-dimensional regression literature: (1) lack of effective approaches to addressing model selection uncertainty and providing valid inference for the underlying true model; (2) lack of effective inference approaches to guaranteeing finite-sample performance. We provide both finite-sample and asymptotic results to theoretically guarantee the performance of the proposed methods. In addition, our numerical results demonstrate that the proposed methods are valid and achieve better coverage with smaller confidence sets than the current state-of-the-art approaches, such as debiasing and bootstrap approaches.

1. Introduction High-dimensional linear regression plays an important role in modern statistics, with applications ranging from signal processing [57] to econometrics [3, 31] to bioinformatics [53]. There has been a large amount of literature on this topic in the past 30 years. The earlier research focused more on estimation/detection problems such as coefficients estimation [8, 50] and support recovery [63, 68]. Starting with the work of debiased Lasso [30, 42, 61], the more difficult task of inference comes to the central stage. Some recent works on inference include confidence interval construction [30, 42, 61], multiple testing of regression coefficients [38], and post-selection inference [34, 48, 51].

Despite many works on the topic, several important open problems remain. First, most existing works focus on the inference for regression coefficients, while the inference for true model (including uncertainty quantification for model selection) in the high-dimensional regression model is mostly absent. This is partly due to the challenges arising from the discrete nature of the model space, which makes the conventional inference tools built for continuous parameters, such as the central limit theorem and bootstrap theorems, inapplicable. Furthermore, all the results in the literature on high-dimensional inference are

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asymptotic, assuming the sample size goes to infinity, and there are no theories concerning the performance of these procedures under finite-sample settings. The performance of these asymptotic procedures is frequently empirically unsatisfactory, especially when sample size is limited. Thus a procedure with guaranteed finite-sample performance is desirable. Finally, the post-selection inference framework attempts to sidestep the problem of model uncertainty by only making conditional inferences for regression coefficients of the predictors selected by a model selection procedure. If some predictor variables are significant but not selected, no inference results are available on these predictors and we may miss some important signals.

To solve the above problems, we develop a repro samples method that quantifies both the uncertainty in model selection and that in estimation of regression coefficients and their functions. Specifically, we provide a comprehensive inferential approach with which we can construct confidence sets for (a) the true model, (b) a single or any collection of regression coefficients, and (c) both the model and regression coefficients jointly. Moreover, the proposed repro samples approach enjoys finite-sample performance guarantees without requiring a large sample size for all of (a)-(c). Although our work focuses primarily on finite-sample performances, we also provide related large-sample results.

Consider the high-dimensional linear regression problem where we observe an $n \times 1$ response vector $\mathbf{y}_{obs}$ with an $n \times p$ design matrix $\mathbf{X}$, where $p \gg n$. Suppose that the observed $\mathbf{y}_{obs}$ relates only to a subset of predictor variables indexed by $\tau_0 \subset \{1, \dots, p\}$ with

$$(1) \quad \mathbf{y}_{obs} = \mathbf{X}\beta_0^{full} + \sigma_0 \mathbf{u}^{rel} = \mathbf{X}_{\tau_0} \beta_0 + \sigma_0 \mathbf{u}^{rel}.$$

Here, $\mathbf{u}^{rel}$ is the realization of the error term, $\theta_0 = (\tau_0, \beta_0, \sigma_0^2)$ are unknown model parameters, $\mathbf{X} = (\mathbf{X}_{\tau_0}, \mathbf{X}_{\tau_0^C})$, $\beta_0^{full} = (\beta_0^\top, \mathbf{0}_{\tau_0^C}^\top)^\top$ and $\tau_0^C = \{1, \dots, p\} \setminus \tau_0$. Corresponding to model (1), there is a random sample (or population) version of data generation model

$$(2) \quad \mathbf{Y} = \mathbf{X}\beta_0^{full} + \sigma_0 \mathbf{U} = \mathbf{X}_{\tau_0} \beta_0 + \sigma_0 \mathbf{U},$$

of which model (1) is a realization. In general, we assume $\mathbb{E}(\mathbf{U}) = \mathbf{0}$, although further conditions may be required for different inference tasks.

To carry out the inference tasks, two inversion techniques are devised in the proposed repro samples method to handle the discrete parameter τ_0 . The first inversion technique, developed in Section 2 and referred to as *Fisher inversion*, aims to obtain a model candidate set that includes the true τ_0 with a high (close to 1) probability using the observed data and a large set of *reproduced* (simulated) artificial error terms. This model candidate set, typically of a reasonable size, is then used to facilitate the constructions of level- α confidence sets and intervals later in Sections 3 and 4. To obtain a level- α model confidence set for τ_0 in Section 3, we use the second inversion technique, referred to as *Fisher-Dempster inversion* for distinction, to overcome the difficulty that the central limit theory does not apply on a discrete parameter space. The Fisher-Dempster inversion technique inverts a level- α Borel set of possible error realizations to get a level- α confidence set for τ_0 . Unlike the conventional Wald-test type of methods, the proposed repro samples method directly provides the desired confidence sets without having to estimate (τ_0, β_0) or any other model parameters.

1.1. *Contributions* To summarize, this paper has the following contributions.

1. We propose a repro samples method to effectively construct model confidence set and quantify model selection uncertainty for the high-dimensional linear regression model. To the best of our knowledge, it is the only computationally efficient approach that provides a performance-guaranteed model confidence set for τ_0 without data splitting or a prior assumed candidate model set.

2. We develop a novel inference procedure for regression coefficients β_0 . Contrary to other existing methods, our approach does not rely on covariance matrix estimation or a consistent model selection procedure. Therefore, it sidesteps potential issues caused by any inaccurate estimation of the covariance matrix or mis-selection of the model.
 3. We propose a novel and efficient way to find candidate models using synthetically generated residuals. Theoretically, we show that this set of candidate models have a high probability to include the true model τ_0 for both Gaussian and non-Gaussian errors, and we also provide an upper bound for the expected size of the set. Numerically, we have shown in our simulation studies that the proposed model candidate set is of reasonable size, and covers τ_0 with a probability close to 1 even in challenging settings. The model candidate set facilitates the inference for both the true model and the regression coefficients. It can also be used for variable screening, providing superior performances to the traditional screening approaches relying on only marginal relationship.
 4. Theoretically, we show that the proposed inference procedures for both the model and the regression coefficients achieve finite-sample coverage guarantees, while most literature on high-dimensional models focuses only on asymptotic properties. To our knowledge, the proposed method is the first approach that guarantees coverage for finite samples. Additionally, our theory suggests a complementary effect between computational power and sample size: one can achieve valid coverage as long as either is sufficiently large.
 5. Theoretically, we do not need to impose the standard conditions that high-dimensional statistics literature typically requires to obtain a consistent estimation, such as the restricted isometry property or restricted eigenvalue conditions [5, 61]. Neither do we need to require signal strength conditions [5, 63], which is usually necessary for consistent model selection. We also provide a discussion on conditions required for different implementations of the proposed procedure under computational considerations.
 6. Finally, through extensive numerical studies, we show that the proposed method produces better performed confidence sets than those of the state-of-the-art debiased Lasso estimators [30, 61]. Because of the finite-sample validity guarantee, our method achieves the desired coverage even in small-sample regimes, while the existing methods can not.
- Overall, we provide a comprehensive framework that subsumes existing inference approaches by two means: we consider a broader set of marginal and joint inference problems to account for uncertainties of estimating both the model and the regression coefficients; and we provide supporting theories to guarantee both finite- and large-sample performances.

1.2. *Related works* There has been much effort in recent years to develop inference procedures for regression coefficients β_0^{full} or functions of β_0^{full} in high-dimensional linear regression models. On the inference for a single coefficient parameter, [30, 52, 61] propose the debiased Lasso estimator and develop its asymptotic distribution. Other works along this line include [9, 11, 12, 15, 16]. Moreover, [6, 17, 42, 62, 64, 66, 67] investigate simultaneous inference on a subset or all of β_0^{full} . Additionally, quadratic and more general functions of β_0^{full} have been studied in [1, 23, 28, 41, 65, 66]. However, all the existing approaches are developed using large-sample theories and do not have any finite-sample performance guarantees. The inference for τ_0 , on the other hand, is almost entirely absent in high-dimensional statistics literature, although there are a few works in low-dimensional setting. [21, 25] construct a model confidence set utilizing sequential testing procedures against a pre-specified finite-dimensional full model, which cannot be well-defined in the high-dimensional setting. [36] proposes a novel concept of model confidence bounds to confine τ_0 within a pair of nested models. However, the method relies on selection consistency and bootstrap validity, and is computationally expensive for high-dimensional data.

A recent work by the authors [59] provides a repro samples framework for statistical inference under a general setup, in which the number of parameters p is less than the number

of observations n . The current paper focuses on the high-dimensional $p \gg n$ case that was not discussed in [59]. New procedures and theoretical results with conditions tailored to high-dimensional models that guarantee the performance of the proposed method in both finite and large-sample cases are developed. Finally, as discussed in [59], the repro samples approach is related to other modern simulation-based procedures, such as the bootstrap [9, 18], the approximate Bayesian computation [2, 13], the inferential models [40] and the generalized fiducial inference [24], where artificial data are used to address inference problems.

1.3. Notation For any $p \in \mathbb{N}^+$, we let $[p] = \{1, \dots, p\}$. For a vector $\mathbf{v}$, we let v_i be the i -th entry. For a set S , let $|S|$ be the cardinality of S . For two positive sequences $\{a_k\}$ and $\{b_k\}$, write $a_k = O(b_k)$, if $\lim_{k \rightarrow \infty} (a_k/b_k) < \infty$; write $a_k = o(b_k)$, if $\lim_{k \rightarrow \infty} (a_k/b_k) = 0$. We use $\mathbb{P}$ for probability and $\mathbb{E}$ for expectation and add subscripts (eg., $\mathbb{P}_{\mathbf{U}}$ and $\mathbb{E}_{\mathbf{U}}$) to indicate source of randomness. We use $\hat{\mathbb{P}}$ and $\hat{\mathbb{E}}$ for empirical probability and expectation. For a $\beta \in \mathbb{R}^p$ and $\tau \subset [p]$, we use β_τ to denote the sub-vector of β , containing the entries of β that are associated with the indices in τ . The model space $\mathcal{M} = 2^{[p]}$. For a matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$, let $\text{span}(\mathbf{M})$ be the vector space spanned by the columns of $\mathbf{M}$: $\text{span}(\mathbf{M}) = \{\mathbf{M}\mathbf{v} : \mathbf{v} \in \mathbb{R}^n\}$. We also call $\mathbf{M}(\mathbf{M}^\top \mathbf{M})^{-1} \mathbf{M}^\top$ the projection matrix of $\mathbf{M}$, and $\mathbf{I}$ is the identity matrix. Lastly, we use $\Gamma_\alpha^\theta(\mathbf{y}_{obs})$, $\Gamma_\alpha^\tau(\mathbf{y}_{obs})$ and $\Gamma_\alpha^{\beta_\Lambda}(\mathbf{y}_{obs})$ to denote the level- α confidence set for θ_0 , τ_0 and $\beta_{0,\Lambda}$, respectively, where Λ is any subset of $[p]$. Here, the superscript θ of $\Gamma_\alpha^\theta(\mathbf{y}_{obs})$ simply indicates the target parameter of the confidence set is θ , and the set $\Gamma_\alpha^\theta(\mathbf{y}_{obs})$ does not depend on any particular value of θ . Notations for other confidence sets, such as $\Gamma_\alpha^\tau(\mathbf{y}_{obs})$, $\Gamma_\alpha^{\beta_\Lambda}(\mathbf{y}_{obs})$, etc., are defined similarly.

Finally, we refer to a simulated copy of artificial $\mathbf{u}^* \sim \mathbf{U}$ as a *repro* copy of the realized $\mathbf{u}^{rel}$ and the artificial data $\mathbf{y}^* = X_\tau \beta_\tau + \sigma \mathbf{u}^*$ as a *repro sample* of $\mathbf{y}^{obs}$ for a potential set of values $(\tau, \beta_\tau, \sigma^2)$. The key of our approach is to study and relate this $\mathbf{u}^*$ with $\mathbf{u}^{rel}$ and the $\mathbf{y}^*$ with $\mathbf{y}^{obs}$. We generally refer to our method, developed by using the copies of $\mathbf{u}^*$ and $\mathbf{y}^*$, as a *repro samples method*. We will provide more details in each of the sections.

1.4. Organization The paper is organized as follows. Section 2 provides a data-driven approach to effectively construct a set of candidate models that will include the true model τ_0 with a high (close to 1) probability. Section 3 utilizes the candidate set to construct a level- α confidence set for τ_0 , and provides both finite-sample and large-sample guarantees. Section 4 studies the inference problems of regression coefficients, including inference for β_0^{full} , linear transformations of β_0^{full} and functions of β_0^{full} . Section 5 provides numerical illustrations of the proposed methods and compares the coverage and size of the constructed confidence sets with the bootstrap and state-of-the-art debiased Lasso methods. In Section 6, we perform a real data analysis. Section 7 concludes the paper with a discussion of our results and future research directions. Theoretical proofs, technical lemmas, as well as additional discussions and numerical results are deferred to Appendices A–F in the supplementary materials [54].

2. Finding candidate models for τ_0 In this section, we propose a novel procedure to efficiently find possible candidate models for τ_0 . In Section 2.1, to rigorously set up our problem and eliminate possible non-identifiability issues, we formally define the target true sparse model τ_0 as the smallest model that generates the data. Section 2.2 introduces an effective computing algorithm and Fisher inversion method to obtain a set of candidate models for τ_0 . Section 2.3 proves that the model candidate set obtained in Section 2.2 is guaranteed to cover τ_0 with a probability close to 1 under the Gaussian error model assumption, in both finite-sample and asymptotic settings. Moreover, we also provide a theoretical upper bound for the size of the model candidate set. In Section 2.4, we show that the finite-sample coverage results also hold when the error term follows a number of other distributions, such as when $\mathbf{U}$ is heterogeneous, Cauchy, t -distributed, contaminated Gaussian or sub-Gaussian distributed. In addition, we show that the large-sample result continues to hold under sub-Gaussian errors.

2.1. *Identifiability and definition of τ_0* Under the high-dimensional setting with $p \gg n$, there might be another model $\tilde{\tau}_0$ and corresponding coefficients $\beta_{\tilde{\tau}_0}$ such that $\mathbf{X}_{\tau_0}\beta_0 = \mathbf{X}_{\tilde{\tau}_0}\beta_{\tilde{\tau}_0}$. Even when we know both $\mathbf{y}_{obs}$ and the realized noise $\mathbf{u}^{rel}$, it is not possible to tell apart τ_0 and $\tilde{\tau}_0$, since $\mathbf{y}_{obs} = \mathbf{X}_{\tau_0}\beta_0 + \sigma_0\mathbf{u}^{rel} = \mathbf{X}_{\tilde{\tau}_0}\beta_{\tilde{\tau}_0} + \sigma_0\mathbf{u}^{rel}$. We refer to this as an *identifiability issue*. To address this issue and uniquely define τ_0 , the conventional practice in the high-dimensional regression literature [e.g., 5, 50] is to favor the smaller model, since as stated in [50], in real applications researchers would often prefer and be interested in the simplest (smallest) model that generates the observed data for better prediction performance and model interpretation. Commonly used penalized regression approaches, such as Lasso [50], SCAD [19], and MCP [60], all employ penalty terms designed to favor smaller models. In this paper, to address this identifiability issue, we follow the same practice to rigorously re-define τ_0 as the smallest model among the set $\{\tau \in \mathcal{M} | \mathbf{X}_\tau\beta_\tau = \mathbf{X}_{\tau_0}\beta_0, \text{ for some } \beta_\tau\}$:

$$(3) \quad \tau'_0 = \underset{\{\tau | \mathbf{X}_\tau\beta_\tau = \mathbf{X}_{\tau_0}\beta_0\}}{\operatorname{argmin}} |\tau|.$$

Throughout the paper, we assume that the true model τ'_0 defined in (3) is unique, which we refer to as the *identifiability condition*. For notational simplicity, we will still refer τ'_0 as τ_0 and $\beta_{\tau'_0}$ as β_0 in the remainder of the paper. Our inference target is this set of (τ_0, β_0) just defined.

Furthermore, we follow [47] to define the degree of separation between model τ_0 and other models of equal or smaller model sizes as

$$C_{\min} = \min_{\{\tau: \tau \neq \tau_0, |\tau| \leq |\tau_0|\}} \frac{1}{n \max(|\tau_0| - |\tau|, 1)} \|\mathbf{X}_{\tau_0}\beta_0 - \mathbf{X}_\tau\beta_\tau\|_2^2.$$

Under the identifiability condition mentioned above, we have $C_{\min} > 0$. The notion $C_{\min}$ is related to $\beta_{\min}$ of the β -min condition in the literature [e.g., 50]. However, unlike the existing literature, we do not impose any assumption on $C_{\min}$ other than that $C_{\min} > 0$.

2.2. *Algorithm for finding candidate models* Here, we use an inversion method to construct a set of candidate models for τ_0 . To illustrate the basis of this inversion method, we first show that we can recover the true model τ_0 in an ideal (unrealistic) case assuming that the realization of the error term $\mathbf{u}^{rel}$ is known. In particular, Lemma 1 below states that τ_0 defined in (3) can be expressed in terms of the given realization $(\mathbf{y}_{obs}, \mathbf{u}^{rel})$ using an optimization statement.

LEMMA 1. *Let $\mathbf{H}_\tau$ be the projection matrix of $\mathbf{X}_\tau$ and $\mathbf{H}_{\tau, \mathbf{u}^{rel}}$ be the projection matrix of $(\mathbf{X}_\tau, \mathbf{u}^{rel})$. Let $\gamma_{(\mathbf{u}^{rel}, \tau_0)}^2 = 1 - \min_{\{\tau: |\tau| < |\tau_0|\}} \frac{\|(I - \mathbf{H}_{\tau, \mathbf{u}^{rel}})\mathbf{X}_{\tau_0}\beta_0\|^2}{\|(I - \mathbf{H}_\tau)\mathbf{X}_{\tau_0}\beta_0\|^2} < 1$. Then, given $\mathbf{u}^{rel}$, τ_0 defined in (3) satisfies*

$$(4) \quad \tau_0 = \underset{\tau}{\operatorname{argmin}} \left\{ \min_{\beta_\tau, \sigma} \left\{ \|\mathbf{y}_{obs} - \mathbf{X}_\tau\beta_\tau - \sigma\mathbf{u}^{rel}\|_2^2 + \lambda|\tau| \right\} \right\},$$

and moreover $(\tau_0, \beta_0, \sigma_0) = \underset{\tau, \beta_\tau, \sigma}{\operatorname{argmin}} \left\{ \|\mathbf{y}_{obs} - \mathbf{X}_\tau\beta_\tau - \sigma\mathbf{u}^{rel}\|_2^2 + \lambda|\tau| \right\}$ for any $0 < \lambda < n[1 - \gamma_{(\mathbf{u}^{rel}, \tau_0)}^2]C_{\min}$.

In practice, however, we do not know the realized errors $\mathbf{u}^{rel}$ so we cannot directly apply Lemma 1. Nonetheless, equation (4) offers guidance on constructing a set of candidate models for τ_0 . More specifically, we generate a large number of, say d , copies of Monte Carlo $\mathbf{u}_1^*, \dots, \mathbf{u}_d^* \stackrel{i.i.d.}{\sim} \mathbf{U}$. Then instead of solving (4) with the realized $\mathbf{u}^{rel}$, we compute

$$(5) \quad \hat{\tau}_b = \underset{\tau}{\operatorname{argmin}} \left\{ \min_{\beta_\tau, \sigma} \left\{ \|\mathbf{y}_{obs} - \mathbf{X}_\tau\beta_\tau - \sigma\mathbf{u}_b^*\|_2^2 + \lambda|\tau| \right\} \right\},$$

for each $\mathbf{u}_b^*, b = 1, \dots, d$. After that, we collect all $\hat{\tau}_b$'s to form a candidate set for τ_0 :

$$(6) \quad S^{(d)} = \{\hat{\tau}_b : b = 1, \dots, d\}.$$

121 Since the mapping function from $\mathbf{u}_b^* \in \mathbb{R}^n$ to $\hat{\tau}_b \in \mathcal{M} = 2^{[p]}$ in (5) is a many-to-one mapping,
 122 many of the d copies of $\hat{\tau}_b$'s obtained by (5) are identical. The size $|S^{(d)}|$ is often much smaller
 123 than d . See Theorem 3 of Section 2.3 for a theoretical result on the size of the candidate set.

124 The only difference between (5) and (4) is that we replace $\mathbf{u}^{rel}$ with $\mathbf{u}_b^*$. Since the mapping
 125 function from $\mathbf{u}_b^*$ to $\hat{\tau}_b$ in (5) is many-to-one, many $\mathbf{u}_b^*$'s that are close to each other map to an
 126 identical $\hat{\tau}_b$. One could imagine that if some $\mathbf{u}_b^*$ is in a neighborhood of $\mathbf{u}^{rel}$, then for such
 127 $\mathbf{u}_b^*$'s, the event $\{\hat{\tau}_b = \tau_0\}$ is very likely to happen. The size of such a neighborhood depends on
 128 the separation metric $C_{\min}$ and the sample size, yet its probability measure is always positive
 129 under the identifiability condition described in Section 2.1. As a result, as long as d , the number
 130 of repro copies, is sufficiently large, eventually some $\mathbf{u}_b^*$ will fall in this neighborhood, leading
 131 to $\hat{\tau}_b = \tau_0$ and hence the candidate set $S^{(d)}$ contains the true model τ_0 . Formal theorems that
 132 support this method for different error distributions are presented in Section 2.3–2.4.

133 To put it succinctly, we summarize the aforementioned procedure in Algorithm 1 below.

Algorithm 1 Search of Candidate Models

Input: Design matrix $\mathbf{X}$, response vector $\mathbf{y}_{obs}$, the number of repro samples d .

Output: Candidate Models $S^{(d)}$.

Step 1: Simulate a large number d copies of $\mathbf{u}^* \sim \mathbf{U} \sim N(0, \mathbf{I}_n)$. Denote the d copies by $\mathbf{u}_b^*, b = 1, \dots, d$.

Step 2: Compute $\hat{\tau}_{b,\lambda} = \underset{(\beta_\tau, \sigma)}{\operatorname{argmin}} [\lambda|\tau| + \|\mathbf{y}_{obs} - \mathbf{X}_\tau \beta_\tau - \sigma \mathbf{u}_b^*\|_2^2]$ for $b = 1, \dots, d$ and a grid of λ
 values. For each b , use certain selection criteria to pick a subset of all values of λ , denoted as Λ_b .

Step 3: Construct $S^{(d)} = \{\hat{\tau}_{b,\lambda} : \lambda \in \Lambda_b, b = 1, \dots, d\}$.

134 **REMARK 1.** [Practical implementation of Algorithm 1] When we implement Algorithm 1,
 135 we need to consider two practical issues: a) how to handle the tuning parameter λ in
 136 the penalty term, and b) solving an optimization problem with a L_0 penalty $\lambda|\tau|$ is often
 137 computationally difficult for high-dimensional data. In our implementation in the numerical
 138 study Sections 5 and 6, we follow common practices in the literature to handle these two issues.
 139 First, it is common to use a selection criterion to determine the value of the tuning parameter
 140 λ [10, 19, 50]. We use the extended BIC (EBIC) [10] to determine λ , due to its good empirical
 141 performance and asymptotic model selection consistency in high-dimensional settings. Second,
 142 solving an optimization problem with the L_0 penalty is computationally expensive and yields
 143 unstable results [39]. In practice, researchers often use a surrogate to replace the L_0 penalty.
 144 In our numerical studies, we adopt the adaptive Lasso [68] as a surrogate for the L_0 penalty
 145 in (5) because of its simplicity and convexity. One may also use other surrogates like the
 146 truncated Lasso penalty [46], smoothly clipped absolute deviation penalty (SCAD) [19], or
 147 the minimax concave penalty (MCP) [60], among others. Although computationally more
 148 efficient, using some of these penalties may require us to make additional assumptions on the
 149 design matrix. See Remark 2 for further discussions. In this paper, we develop our general
 150 theories using the L_0 penalty and constraint rather than a specific surrogate penalty function,
 151 since we would like to understand the fundamental properties and allow for the flexibility of
 152 using any penalty or constraint within the proposed repro samples framework.

153 Equations (4) and (5) are inversion operations that solve for τ when given $\mathbf{y}_{obs}$ and the error
 154 term $\mathbf{u}$. The difference is that (4) assumes the realized $\mathbf{u}^{rel}$ is known while (5) uses a simulated
 155 $\mathbf{u}_b^*$. This technique of using a random $\mathbf{u}_b^*$ to replace $\mathbf{u}^{rel}$ in an inversion can be traced back
 156 to Fisher's fiducial inference [24, 49]. Therefore, we refer to the inversion method used in
 157 Algorithm 1 as *Fisher inversion*. Here, we use it to assemble potential candidate models for
 158 τ_0 , which reduces the size of the effective model space from 2^p to $|S^{(d)}|$. In Section 3, we will
 159 develop a different inversion technique to construct the level- α confidence set for τ_0 .

2.3. *Theoretical results for models with Gaussian errors* In this subsection, we present theoretical guarantees of our method under Gaussian error $\mathbf{U} \sim N(\mathbf{0}, \mathbf{I}_n)$, and extend the results beyond the Gaussian error model in the next subsection. Here, we show in Theorems 1-2 that $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \rightarrow 0$, in the following two cases: 1) the sample size n is finite and $d \rightarrow \infty$, 2) d is finite and $n \rightarrow \infty$, respectively. The probability $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\cdot)$ refers to the joint distribution of $\mathbf{U}$ and $\mathcal{U}^d = (\mathbf{U}_1^*, \dots, \mathbf{U}_d^*)$, where $\mathbf{U}_b^*$ is a Monte Carlo copy of $N(\mathbf{0}, \mathbf{I}_n)$.

THEOREM 1. *Suppose $n - |\tau_0| > 4$. For any $\delta > 0$, there exists a constant $\gamma_\delta \in (0, 1)$ such that when $\lambda \in [4n\gamma_\delta^{1/2} \{2 + 2(|\tau_0| + 1) \frac{\log(p/2)}{n}\} \sigma_0^2, \frac{n\gamma_\delta^{1/4}}{6} C_{\min}]$, the finite-sample probability bound that the true model is not included in the model candidates set $S^{(d)}$, obtained by (6) with the objective function (5), is as follows,*

$$(7) \quad \mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \leq \left(1 - \frac{\gamma_\delta^{n-1}}{n-1}\right)^d + \delta.$$

Therefore as $d \rightarrow \infty$, $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \rightarrow \delta$, where $\delta > 0$ is arbitrarily small.

THEOREM 2. *Suppose $\frac{\lambda}{n} \in [6\sigma_0^2 \frac{(|\tau_0|+1)\log(p/2)}{n} + t, 0.05C_{\min}]$ for a positive constant $t > 0$. Then the finite-sample probability bound that the true model is not included in the confidence set $S^{(d)}$, obtained by (6) with the objective function (5) for any finite d is as follows,*

$$(8) \quad \begin{aligned} \mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \leq & 6 \exp \left[-\frac{n}{18\sigma_0^2} \left\{ 0.3C_{\min} - 36 \frac{\log(p+1)}{n} \sigma_0^2 \right\} \right] + 3 \exp \left(-\frac{nt}{3\sigma_0^2} \right) \\ & + \exp \left\{ -nd \left(0.23 - \frac{|\tau_0| \log(p) + 2}{n} \right) \right\}, \end{aligned}$$

Therefore $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \rightarrow 0$ for any d as $n \rightarrow \infty$, if $\frac{|\tau_0| \log(p)}{n} < 0.23$ and $C_{\min} > 120 \frac{(|\tau_0|+2)\log(p+1)}{n} \sigma_0^2$ when n is large enough.

The two theorems above suggest two complementary driving forces of the coverage validity: the sample size and the computation time measured by d . In cases when the sample is limited, Theorem 1 implies that we can recover the signal with a valid coverage as long as the computation time (linearly scaled with d) goes to infinity; in cases when the computational resources are limited, Theorem 2 then indicates collecting sufficient samples will result in a valid coverage guarantee. In Theorem 1, for any finite n, p , the lower bound for λ is of the same order as $\gamma_\delta^{1/2}$, and the upper bound is of the same order as $\gamma_\delta^{1/4}$. Therefore the range of λ always exists for a γ_δ that is small enough. In Theorem 2, the existence of the range of λ follows from $C_{\min} > 120 \frac{(|\tau_0|+2)\log(p+1)}{n} \sigma_0^2$ when n is large enough. Therefore, in both theorems, the required range for λ is a non-empty interval of positive length, although this interval is smaller for a smaller $C_{\min}$.

REMARK 2. *In this paper, we develop our general theorems using the L_0 penalty or constraints, rather than any specific version of surrogates, to keep the theory general and allow researchers to select the surrogate that best suits their needs. In our implementation in the paper, we used the adaptive Lasso penalty as a surrogate for the L_0 penalty since it is computationally efficient and performs comparably to commonly used non-convex penalties. To obtain similar result of Theorem 1 tailored specifically for the adaptive Lasso penalty, we would need impose an additional condition called the minimum adaptive restrictive eigenvalue condition [5, Ch 6& 7]. This condition is similar, but slightly weaker than the restricted strong convexity and it is also weaker than the irrepresentable condition [5, Ch 6& 7]. Moreover, the simulation results of Model (M5) in Section 5 suggest that our current implementation of the repro samples approach still performs well empirically even when the minimum adaptive*

restricted eigenvalue condition required for the adaptive Lasso does not hold. Besides the adaptive Lasso, there are other possible surrogates (e.g., Lasso, adaptive Lasso, SCAD, MCP, etc.) for the L_0 penalty. Whether we need additional conditions on the design matrix and what these conditions are depend on the specific L_0 surrogate we use in our implementation. In general, there is a trade-off between additional conditions required and computational cost. For example, if we choose to adopt the truncated Lasso penalty (TLP) proposed by [47], then no additional condition is required on the design matrix. Alternatively, we can choose to use a constrained least squares approach as opposed to the penalized approach (see Appendix A in [54] for the formulation and theories regarding the constraint approach). In this case, if we choose to use the constrained L_0 regression to estimate the models, which we can achieve with the modern mixed integer optimization approach [4], we would not need any condition on the design matrix either. However, both of these approaches demand substantially higher computational cost compared to a convex penalty function like the adaptive Lasso.

Besides the coverage results above, another important aspect is the size of the candidate set $|S^{(d)}|$. Theorem 3 below provides a theoretical bound for the expectation of the size of the model candidate set $\mathbb{E}(|S^{(d)}|)$. In the theorem, for any model τ with $|\tau| \leq |\tau_0|$, we define the model distance between τ and the truth τ_0 as $C_\tau = \frac{1}{n \max(|\tau_0|, 1)} \|\mathbf{X}_{\tau_0} \beta_0 - \mathbf{X}_\tau \beta_\tau\|_2^2$.

THEOREM 3. *Let $\Xi(c) = \{\tau : |\tau| \leq |\tau_0|, C_\tau \leq c\}$ be the set of τ smaller than τ_0 that are close to τ_0 . Then for $\frac{\lambda}{n} \geq \frac{3\sigma_0^2(|\tau_0|+1)\{\log(p-|\tau_0|)+\log(|\tau_0|)+\frac{2}{3}\}}{n} + t$, where $t > 0$,*

$$\mathbb{E}(|S^{(d)}|) \leq |\Xi(\bar{c})| + \sum_{\{\tau: |\tau| \leq |\tau_0|, C_\tau > \bar{c}\}} \exp\left\{-\frac{n}{20\sigma^2}(C_\tau - \bar{c})\right\} + 3 \exp\left(-\frac{nt}{3\sigma_0^2}\right),$$

where $\bar{c} = \{7\lambda + 14\sigma_0^2(1 + 1.5 \log d)\}/n = O(\log(d)/n)$, where d is the number of repro samples used in Algorithm 1.

Intuitively, if an alternative model τ is closer to τ_0 in that C_τ is small, it should be more likely to be included in the model candidate set $S^{(d)}$. Therefore the candidate set would include models that are close to τ_0 , and models that are farther away from τ_0 would be included with a smaller probability. As a result, the size of the candidate set depends on (a) how many models are close to τ_0 , (b) the probability of other models included. This intuition is verified explicitly by the result in Theorem 3. Specifically, if $\log(p)/n = o(1)$, then with high probability, the candidate set will include all the models with $C_\tau = o(\log(d)/n)$, where d is the number of repro copies in Algorithm 1, and it will include those with $C_\tau = O(\log(d)/n)$ with a positive probability. The larger the C_τ is, the smaller the probability τ being included. Moreover, a larger sample size leads to smaller model selection uncertainty, typically resulting in a smaller candidate set, aligning with our expectations. Additionally, the impact of the repro sample size d on the size of $S^{(d)}$ is logarithmic. Finally, we see from Theorem 3 that the contribution to the cardinality of the model candidate set from all models larger than τ_0 is bounded by $3 \exp\left(-\frac{nt}{3\sigma_0^2}\right)$, therefore the candidate models include only models of size similar or smaller than τ_0 with probability close to 1 due to the regularization in Step 2 of Algorithm 1.

2.4. Heterogeneous, non-Gaussian and sub-Gaussian error models In this subsection, we show that even when the model error assumption $\mathbf{U} \sim N(0, \mathbf{I}_n)$ is violated, the model candidate set $S^{(d)}$ obtained using Algorithm 1 with $\mathbf{u}_b^* \sim N(0, \mathbf{I}_n)$ can still cover the true model τ_0 with a high probability.

We first show in Theorem 4 below that the results in Theorem 1 still hold when the linear model in (2) is now generalized to the following,

$$(9) \quad \mathbf{Y} = \mathbf{X} \beta_0^{full} + \sigma_0 \mathbf{U}_\Omega = \mathbf{X}_{\tau_0} \beta_0 + \sigma_0 \mathbf{U}_\Omega.$$

Here, the error term $\mathbf{U}_\Omega = \text{diag}(\boldsymbol{\Omega})\mathbf{U}$, $\mathbf{U} \sim N(0, \mathbf{I}_n)$ and $\boldsymbol{\Omega} = (\Omega_1, \dots, \Omega_n)$ is an $n \times 1$ fixed vector with each $\Omega_i = O(1)$, $1 \leq i \leq n$, or a random vector independent of $\mathbf{U}$, with each $\Omega_i = O_p(1)$, $1 \leq i \leq n$.

THEOREM 4. Suppose $\mathbf{Y}$ is generated by (9), $n - |\tau_0| > 4$, $|\{\Omega_i : \Omega_i > 0\}| > |\tau_0|$. Then for any $\delta > 0$, there exists a constant $\gamma_\delta \in (0, 1)$ such that when $\lambda \in [n\gamma_\delta^{1/2}\{2 + 2(|\tau_0| + 1)\frac{\log(p/2)}{n}\}, n\gamma_\delta^{1/4}\frac{C_{\min}}{6}]$, the finite-sample probability bound that the true model is not included in the model candidates set $S^{(d)}$, obtained by (6) with the objective function (5), is as follows,

$$\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \leq \left(1 - \frac{\gamma_\delta^{n-1}}{n-1}\right)^d + \delta.$$

Therefore as $d \rightarrow \infty$, $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \rightarrow \delta$, where $\delta > 0$ is arbitrarily small.

The above theorem shows that when the error term is generalized from $\mathbf{U}$ to $\mathbf{U}_\Omega$, the candidate set $S^{(d)}$ still cover the truth τ_0 with an arbitrarily high probability, as long as d is large enough. The generalized error term $\mathbf{U}_\Omega$ covers a wide range of non-Gaussian error models, including heterogeneous variances, Cauchy distribution, t -distribution, contaminated Gaussian distribution, etc. Below, we discuss how these non-Gaussian errors are connected with $\mathbf{U}_\Omega$.

Heterogeneous Variance. In this case, $\mathbf{U}_\Omega \sim N(0, \Sigma)$, where $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_2^2)$. Therefore we can just make Ω fixed such that $P(\Omega_i = \sigma_i) = 1$.

Cauchy Distribution. When the error is Cauchy distribution, we can simply make $\Omega_i = 1/|Z_i|$, where Z_i are i.i.d. $N(0, 1)$.

T-distribution with degree of freedoms ν . Since random variables with t -distribution can be formulated by a ratio of a normal random variable and the square root of a Chi-square random variable divided by its degree of freedom, we make $\Omega_i \sim \frac{1}{\sqrt{\chi_\nu^2/\nu}}$.

Contaminated Gaussian Distribution. In cases where some observations are contaminated, leading to a larger variance for the contaminated samples, we can make $\Omega_i = (1 - Z_i) + wZ_i$, where w is a constant, usually larger than 1, and $Z_i, i = 1, \dots, n$ are i.i.d. $\text{Bernoulli}(p)$. Here p represents the proportions of contaminated distributions.

Combination of the above. This is when Ω_i follows a mixture of the above distributions.

Sub-Gaussian error models are another set of non-Gaussian error models used in high-dimensional settings due to their flexibility and robustness. We show that the results of Theorem 1 and Theorem 2 still hold for the sub-Gaussian error models, and thus candidate set $S^{(d)}$ obtained in Algorithm 1 can still cover the true τ_0 with a high probability.

THEOREM 5. Suppose $\mathbf{Y}$ is generated by (2) with $\mathbf{U}$ being a sub-Gaussian vector with sub-Gaussian norms bounded by a universal constant, and $n - |\tau_0| > 4$. For any $\delta > 0$, there exists a constant $\gamma_\delta \in (0, 1)$ and $\zeta_\delta > 0$ such that when $\lambda \in [n\gamma_\delta^{1/2}\{2 + 2(|\tau_0| + 1)\frac{\log(p/2)}{n}\}, n\gamma_\delta^{1/4}\frac{C_{\min}}{6}]$, the finite-sample probability bound that the true model is not included in the model candidates set $S^{(d)}$, obtained by (6) with the objective function (5), is as follows,

$$(10) \quad \mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \leq e^{-\zeta_\delta d} + \delta.$$

Therefore as $d \rightarrow \infty$, $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \rightarrow \delta$, where $\delta > 0$ is arbitrarily small.

For the model candidate set $S^{(d)}$ in both theorems, the repro errors used in Algorithm 1 are still sampled from $\mathbf{u}_b^* \sim N(0, \mathbf{I}_n)$, even though the true underlying $\mathbf{u}^{rel}$ is not. This is possible because $\mathbf{u}^{rel}$ is a vector in $\mathbb{R}^n$, as long as $\mathbf{u}^{rel}$ is not too extreme, we often can find some $\mathbf{u}_b^* \sim N(0, \mathbf{I}_n)$ in its neighborhood when $d \rightarrow \infty$. Under the non-Gaussian settings considered in Theorems 4 and 5, we are able to quantify such a neighborhood of $\mathbf{u}^{rel}$ that also maps $\mathbf{u}_b^*$ to τ_0 . A nice implication of these results is that we do not need to know exactly the error distribution of the model, as long as it is one of those in Theorems 4 and 5, the model candidate

set $S^{(d)}$ obtained by Algorithm 1 contains the true τ_0 with a high probability. Furthermore, we later extend the finite-sample result here to arbitrary error distributions with finite second moments; see the Discussion section.

THEOREM 6. *Suppose $\mathbf{Y}$ is generated by (2) with $\mathbf{U}$ being a sub-Gaussian vector with sub-Gaussian norms bounded by a universal constant and $\frac{\lambda}{n} \in [6\sigma_0^2 \frac{(|\tau_0|+1)\log(p/2)}{n} + t, 0.05C_{\min}]$ for a positive constant $t > 0$. Then the probability bound that the true model is not included in the confidence set $S^{(d)}$, obtained by (6) with the objective function (5) for any finite d is as follows,*

$$\begin{aligned} \mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) &\leq \exp \left[-\frac{n}{18\sigma_0^2} \left\{ 0.3C_{\min} - 36 \frac{\log(p+1)}{n} \sigma_0^2 \right\} \right] + 3 \exp \left(-\frac{nt}{3\sigma_0^2} \right) \\ &\quad + \exp \left\{ -nd \left(0.23 - \frac{|\tau_0| \log(p) + 2}{n} \right) \right\}, \end{aligned}$$

Therefore $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}(\tau_0 \notin S^{(d)}) \rightarrow 0$ for any d as $n \rightarrow \infty$, if $\frac{|\tau_0| \log(p)}{n} < 0.23$ and $C_{\min} > 120 \frac{(|\tau_0|+2)\log(p+1)}{n} \sigma_0^2$ when n is large enough.

The above theorem extends the result in Theorem 2 to models with sub-Gaussian errors. It indicates that when sample size is large, we can recover the truth τ_0 with Algorithm 1 with a limited number of repro samples for models with sub-Gaussian errors. Similar to the implications of Theorem 4 and 5, here we do not need to know the distribution of $\mathbf{U}$, and only require $\mathbf{U}$ to be sub-Gaussian.

3. Construction of a level- α Model Confidence Set In this section, we construct a level- α confidence set for model τ_0 by developing a conditional repro samples method tailored to the problem. Here, we assume $\mathbf{U} \sim N(0, \mathbf{I}_n)$.

For the ease of presenting the general idea of the repro samples method as described in [59], let us first assume that we are interested in making a joint inference about $\boldsymbol{\theta}_0 = (\tau_0, \boldsymbol{\beta}_0^\top, \sigma_0)^\top$ and describe how the method proceeds in this case. The idea is that given any possible value of the parameters $\boldsymbol{\theta} = (\tau, \boldsymbol{\beta}_\tau^\top, \sigma)^\top$, we create an artificial repro sample data $\mathbf{y}^* = \mathbf{X}_\tau \boldsymbol{\beta}_\tau + \sigma \mathbf{u}^*$, where $\mathbf{u}^* \sim N(0, \mathbf{I}_n)$. If $\mathbf{u}^*$ is close to $\mathbf{u}^{rel}$ and $\boldsymbol{\theta}$ is equal or close to $\boldsymbol{\theta}_0$, then we expect $\mathbf{y}^*$ and $\mathbf{y}_{obs}$ to be equal or close. Inversely, for a given value $\boldsymbol{\theta}$, if we can find a $\mathbf{u}^*$ likely matching $\mathbf{u}^{rel}$ such that $\mathbf{y}^*$ matches $\mathbf{y}_{obs}$ (i.e., $\mathbf{y}_{obs} \approx \mathbf{y}^*$), then we cannot rule out that this $\boldsymbol{\theta}$ is a potential value of $\boldsymbol{\theta}_0$. Mathematically, we define

$$(11) \quad \Gamma_\alpha^\theta(\mathbf{y}_{obs}) = \{\boldsymbol{\theta} : \exists \mathbf{u}^* \text{ s.t. } \mathbf{y}_{obs} = G(\boldsymbol{\theta}, \mathbf{u}^*), T(\mathbf{u}^*, \boldsymbol{\theta}) \in B_\alpha(\boldsymbol{\theta})\}.$$

Here, the function $T(\cdot, \cdot)$ is referred to as a *nuclear mapping function* and the set $B_\alpha(\boldsymbol{\theta})$ is a fixed level- α Borel set in $\mathbb{R}^d$ such that

$$(12) \quad \mathbb{P}_{\mathbf{U}}(T(\mathbf{U}, \boldsymbol{\theta}) \in B_\alpha(\boldsymbol{\theta})) \geq \alpha.$$

Again, we clarify that the $\boldsymbol{\theta}$ in the superscript of Γ simply indicates the target parameter is $\boldsymbol{\theta}$ and $\Gamma_\alpha^\theta(\mathbf{y}_{obs})$ in (11) does not rely on any particular value of $\boldsymbol{\theta}$.

The repro samples method uses $\mathbb{P}_{\mathbf{U}}(T(\mathbf{U}, \boldsymbol{\theta}) \in B_\alpha(\boldsymbol{\theta}))$, for each given value $\boldsymbol{\theta}$, as a way to quantify the uncertainty of $\mathbf{U}$ thus also the uncertainty of $\mathbf{Y}$. Moreover, for any nuclear mapping function $T(\mathbf{U}, \boldsymbol{\theta})$, as long as we have a set $B_\alpha(\boldsymbol{\theta})$ such that (12) holds, we can show that the set $\Gamma_\alpha(\mathbf{y}_{obs})$ in (11) is a level- α confidence set [59]. Here, the role of $T(\mathbf{U}, \boldsymbol{\theta})$ under the repro samples framework is similar to that of a test statistic under the classical (Neyman-Pearson) hypothesis testing framework. Besides, a good choice for $T(\mathbf{U}, \boldsymbol{\theta})$ is problem-specific. Effectively, the operation in (11) can be considered as an inversion operation that maps a set of $\mathbf{u}^* \in \mathbb{R}^n$ to a set of $\boldsymbol{\theta} \in \Theta$. Such a mapping is a key element of the Dempster-Shafer calculus [40, 45]. To distinguish the Fisher inversion method introduced in Section 2

that produces a model candidate set, we refer to the techniques used in this section to produce a level- α confidence set for τ_0 as *Fisher-Dempster inversion*.

Our goal in this section to make inference only for the true model τ_0 with $(\beta_\tau^\top, \sigma)$ being the unknown nuisance parameters. First, we write $\mathbf{y}_\theta = \mathbf{X}_\tau \beta_\tau + \sigma \mathbf{u}$, for a $\mathbf{u} \sim \mathbf{U}$. This $\mathbf{y}_\theta$ is a copy of artificial data generated from a given set of parameters $\boldsymbol{\theta} = (\tau, \beta_\tau^\top, \sigma)^\top$. The corresponding random version is

$$(13) \quad \mathbf{Y}_\theta = \mathbf{X}_\tau \beta_\tau + \sigma \mathbf{U}.$$

Then based on the artificial repro sample data $(\mathbf{X}, \mathbf{y}_\theta)$, one can obtain an estimate of τ , denoted by $\hat{\tau}(\mathbf{y}_\theta)$. In this paper, we use

$$(14) \quad \hat{\tau}(\mathbf{y}_\theta) = \arg \min_{\tilde{\tau} \in \mathcal{M}, \beta_{\tilde{\tau}} \in \mathbb{R}^{|\mathcal{T}|}} \|\mathbf{y}_\theta - \mathbf{X}_{\tilde{\tau}} \beta_{\tilde{\tau}}\|^2 \text{ s.t. } |\tilde{\tau}| \leq |\tau|,$$

although in principle we can choose to use another estimator of reasonable performance. We thereafter use (14) to define the nuclear mapping function as $T(\mathbf{u}, \boldsymbol{\theta}) = \hat{\tau}(\mathbf{y}_\theta) = \tilde{T}(\mathbf{y}_\theta, \tau)$. Then we need to find a Borel set $B_\alpha(\boldsymbol{\theta})$ that satisfies (12), i.e., $\mathbb{P}_{\mathbf{U}}(T(\mathbf{U}, \boldsymbol{\theta}) \in B_\alpha(\boldsymbol{\theta})) = \mathbb{P}_{\mathbf{U}}(\tilde{T}(\mathbf{Y}_\theta, \tau) \in B_\alpha(\boldsymbol{\theta})) \geq \alpha$, for the nuclear mapping $\tilde{T}(\mathbf{y}_\theta, \tau)$ defined above. However, the distribution of $\tilde{T}(\mathbf{y}_\theta, \tau)$ involves all of the parameters $\boldsymbol{\theta} = (\tau, \beta_\tau^\top, \sigma)^\top$, including the nuisance parameters β_τ and σ^2 . Therefore, directly obtaining $B_\alpha(\boldsymbol{\theta})$ for all values of $\boldsymbol{\theta}$ is computationally challenging, if not infeasible.

Since it is not feasible to search through the entire space of $(\beta_\tau^\top, \sigma)$, we introduce below an effective conditional repro samples method to handle the nuisance parameters and construct a level- α confidence set for τ_0 . The idea is to first find a quantity $W(\mathbf{U}, \boldsymbol{\theta})$, such that the conditional distribution of the nuclear statistic $T(\mathbf{U}, \boldsymbol{\theta})$ given $W(\mathbf{U}, \boldsymbol{\theta}) = \mathbf{w}$ is free of the nuisance parameters (β, σ) . For now, assume we have such $W(\mathbf{U}, \boldsymbol{\theta})$; we will discuss how to obtain $W(\mathbf{U}, \boldsymbol{\theta})$ for our purpose later in the section. Then, based on the conditional distribution of $T(\mathbf{U}, \boldsymbol{\theta})|W(\mathbf{U}, \boldsymbol{\theta})$, we construct a Borel set $B_\alpha(\tau, \mathbf{w})$ that depends on $\mathbf{w}$, the value of the random quantity $\mathbf{W}(\mathbf{U}, \boldsymbol{\theta})$, but not on (β, σ) , such that

$$(15) \quad \begin{aligned} & \mathbb{P}_{\mathbf{U}|W}\{T(\mathbf{U}, \boldsymbol{\theta}) \in B_\alpha(\tau, \mathbf{w})|W(\mathbf{U}, \boldsymbol{\theta}) = \mathbf{w}\} \\ &= \mathbb{P}_{\mathbf{U}|W}\{\tilde{T}(\mathbf{Y}_\theta, \tau) \in B_\alpha(\tau, \mathbf{w})|W(\mathbf{U}, \boldsymbol{\theta}) = \mathbf{w}\} \geq \alpha. \end{aligned}$$

Accordingly, the marginal probability $\mathbb{P}_{\mathbf{U}}(T(\mathbf{U}, \boldsymbol{\theta}) \in B_\alpha(\tau, W(\mathbf{U}, \boldsymbol{\theta}))) \geq \alpha$.

Now, instead of directly following (11), we construct a subset in the model space $\mathcal{M}$:

$$(16) \quad \begin{aligned} \Gamma_\alpha^\tau(\mathbf{y}_{obs}) &= \{\tau \in \mathcal{M} : \exists \mathbf{u}^* \text{ and } (\beta_\tau, \sigma) \text{ s.t. } \mathbf{y}_{obs} = \mathbf{X}_\tau \beta_\tau + \sigma \mathbf{u}^*, \\ & T(\mathbf{u}^*, \boldsymbol{\theta}) \in B_\alpha(\tau, W(\mathbf{u}^*, \boldsymbol{\theta})), \boldsymbol{\theta} = (\tau, \beta_\tau, \sigma)\}. \end{aligned}$$

The following theorem suggests that $\Gamma_\alpha^\tau(\mathbf{y}_{obs})$ constructed above is a level- α confidence set for the true model τ_0 .

THEOREM 7. *Suppose the conditional distribution of $T(\mathbf{U}, \boldsymbol{\theta})$ given $W(\mathbf{U}, \boldsymbol{\theta}) = \mathbf{w}$ is free of (β_τ, σ) and the Borel set $B_\alpha(\tau, \mathbf{w})$ satisfies (15), then $\mathbb{P}(\tau_0 \in \Gamma_\alpha^\tau(\mathbf{Y})) \geq \alpha$, where the confidence set $\Gamma_\alpha^\tau(\mathbf{Y})$ is defined by (16).*

The remaining task is to find the random quantity $\mathbf{W}(\mathbf{U}, \boldsymbol{\theta})$ and a Borel set $B_\alpha(\tau, \mathbf{w})$ such that the conditional distribution of $T(\mathbf{U}, \boldsymbol{\theta})|W(\mathbf{U}, \boldsymbol{\theta}) = \mathbf{w}$ is free of the nuisance parameters (β, σ) and the inequality (15) holds. Note that we can rewrite (13) as $\mathbf{Y}_\theta = \mathbf{H}_\tau \mathbf{Y}_\theta + (\mathbf{I} - \mathbf{H}_\tau) \mathbf{Y}_\theta = \mathbf{H}_\tau \mathbf{Y}_\theta + \sigma(\mathbf{I} - \mathbf{H}_\tau) \mathbf{U}$, where $\mathbf{H}_\tau = \mathbf{X}_\tau (\mathbf{X}_\tau^\top \mathbf{X}_\tau)^{-1} \mathbf{X}_\tau^\top$ is the projection matrix of $\mathbf{X}_\tau$. Write $\mathbf{A}_\theta(\mathbf{U}) = \mathbf{H}_\tau \mathbf{Y}_\theta = \tilde{\mathbf{A}}_\theta(\mathbf{Y}_\theta)$ and $b_\theta(\mathbf{U}) = \|(\mathbf{I} - \mathbf{H}_\tau) \mathbf{Y}_\theta\| = \tilde{b}_\theta(\mathbf{Y}_\theta)$. We have

$$(17) \quad \mathbf{Y}_\theta = \mathbf{A}_\theta(\mathbf{U}) + b_\theta(\mathbf{U}) \frac{(\mathbf{I} - \mathbf{H}_\tau) \mathbf{U}}{\|(\mathbf{I} - \mathbf{H}_\tau) \mathbf{U}\|} = \tilde{\mathbf{A}}_\theta(\mathbf{Y}_\theta) + \tilde{b}_\theta(\mathbf{Y}_\theta) \frac{(\mathbf{I} - \mathbf{H}_\tau) \mathbf{U}}{\|(\mathbf{I} - \mathbf{H}_\tau) \mathbf{U}\|}.$$

In this equation, the “randomness” of $\mathbf{U}$ (and also $\mathbf{Y}_\theta$) are decomposed into three components, $\mathbf{A}_\theta(\mathbf{U}), b_\theta(\mathbf{U})$ and $(\mathbf{I} - \mathbf{H}_\tau)\mathbf{U}/\|(\mathbf{I} - \mathbf{H}_\tau)\mathbf{U}\|$. Under (13), $(\mathbf{A}_\theta(\mathbf{U}), b_\theta(\mathbf{U})) = (\tilde{\mathbf{A}}_\theta(\mathbf{Y}_\theta), \tilde{b}_\theta(\mathbf{Y}_\theta))$ is a sufficient statistic and the last piece $(\mathbf{I} - \mathbf{H}_\tau)\mathbf{U}/\|(\mathbf{I} - \mathbf{H}_\tau)\mathbf{U}\|$ is an ancillary statistic that is free of the nuisance parameters (β_τ, σ^2) . Based on this partition, we define $\mathbf{W}(\mathbf{U}, \theta) = (\mathbf{A}_\theta(\mathbf{U}), b_\theta(\mathbf{U})) = (\tilde{\mathbf{A}}_\theta(\mathbf{Y}_\theta), \tilde{b}_\theta(\mathbf{Y}_\theta)) = \tilde{\mathbf{W}}(\mathbf{Y}_\theta, \theta)$. It then follows immediately that the conditional distribution of $\mathbf{Y}_\theta \mid \mathbf{W}(\mathbf{U}, \theta) = \mathbf{w}$ is free of (β_τ, σ^2) , so is the conditional probability mass function of $\tilde{T}(\mathbf{Y}_\theta, \tau)$ given $\mathbf{W}(\mathbf{U}, \theta) = \mathbf{w}$,

$$(18) \quad p_{(\mathbf{w}, \tau)}(\tau') = \mathbb{P}_{\mathbf{U}|\mathbf{W}}\{\tilde{T}(\mathbf{Y}_\theta, \tau) = \tau' \mid \mathbf{W}(\mathbf{U}, \theta) = \mathbf{w}\},$$

for any $\tau' \in \mathcal{M}$. Note that, when given $(\mathbf{w}, \tau)$, we can use the model equation (13) to generate many copies of $\mathbf{Y}_\theta$ by repeated draws from $\mathbf{U}$. Therefore we can obtain the conditional probability mass function in (18) through a Monte-Carlo method.

We therefore define the Borel set $B_\alpha(\tau, \mathbf{w})$ as

$$(19) \quad B_\alpha(\tau, \mathbf{w}) = \left\{ \tau^* \in \mathcal{M} : \sum_{\{\tau' : p_{(\mathbf{w}, \tau)}(\tau') \leq p_{(\mathbf{w}, \tau)}(\tau^*)\}} p_{(\mathbf{w}, \tau)}(\tau') \geq 1 - \alpha \right\}.$$

In the proof of Theorem 7 in Appendix C [54], we prove that the conditional probability

$$(20) \quad \mathbb{P}_{\mathbf{Y}_\theta|\mathbf{W}}\{\tilde{T}(\mathbf{Y}_\theta, \tau) \in B_\alpha(\tau, \mathbf{w}) \mid \tilde{\mathbf{W}}(\mathbf{Y}_\theta, \tau) = \mathbf{w}\} \geq \alpha.$$

It follows that, marginally, $\mathbb{P}_{\mathbf{Y}_\theta}\{\tilde{T}(\mathbf{Y}_\theta, \tau) \in B_\alpha(\tau, \tilde{\mathbf{W}}(\mathbf{Y}_\theta, \tau))\} \geq \alpha$. Then by (16) and using the candidate set $S^{(d)}$, we propose the following confidence set for τ_0 ,

$$(21) \quad \bar{\Gamma}_\alpha^\tau(\mathbf{y}_{obs}) = \Gamma_\alpha^\tau(\mathbf{y}_{obs}) \cap S^{(d)} = \left\{ \tau \in S^{(d)} : \tilde{T}(\mathbf{y}_{obs}, \tau) \in B_\alpha\left(\tau, \tilde{\mathbf{W}}(\mathbf{y}_{obs}, \tau)\right) \right\}.$$

To obtain the above confidence set, we use a Monte-Carlo method to compute the conditional probability in (18). We summarize the procedure of constructing the above model confidence set in Algorithm 2, with the size of the Monte-Carlo simulations J .

Theorem 8 below states that $\bar{\Gamma}_\alpha^\tau(\mathbf{y}_{obs})$ in (21) is a level- α confidence set for τ_0 with a guaranteed finite-sample coverage rate, as long as d , the number of repro samples used to construct the candidate set $S^{(d)}$ in Algorithm 1, is large. Theorem 9 states that even when d is limited, $\bar{\Gamma}_\alpha^\tau(\mathbf{y}_{obs})$ is still a level- α confidence set for τ_0 if we have a large sample size n .

THEOREM 8. *Under the conditions in Theorem 1, for any finite n and p , and arbitrarily small $\delta > 0$, the coverage probability of model confidence set $\bar{\Gamma}_\alpha^\tau(\mathbf{y}_{obs})$ constructed above is $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}\{\tau_0 \in \bar{\Gamma}_\alpha^\tau(\mathbf{Y})\} \geq \alpha - \delta - o(e^{-c_1 d})$ as $d \rightarrow \infty$ for some $c_1 > 0$. Further $\mathbb{P}_{\mathbf{Y}|\mathcal{U}^d}\{\tau_0 \in \bar{\Gamma}_\alpha^\tau(\mathbf{Y})\} \geq \alpha - \delta - o_p(e^{-c_1 d})$.*

THEOREM 9. *Under the conditions in Theorem 2, for any finite d , the coverage probability of model confidence set $\bar{\Gamma}_\alpha^\tau(\mathbf{y}_{obs})$ constructed above is $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}\{\tau_0 \in \bar{\Gamma}_\alpha^\tau(\mathbf{Y})\} \geq \alpha - o(e^{-c_2 n})$ as $n \rightarrow \infty$ for some $c_2 > 0$. Further $\mathbb{P}_{\mathbf{Y}|\mathcal{U}^d}\{\tau_0 \in \bar{\Gamma}_\alpha^\tau(\mathbf{Y})\} \geq \alpha - o_p(e^{-c_2 n})$.*

REMARK 3. *When the error $\mathbf{U}$ is non-Gaussian, the theoretical results in Section 2.4 show that the model candidate set $S^{(d)}$ itself, constructed in Algorithm 1, is a valid but conservative confidence set for τ_0 , even without knowing the distribution of $\mathbf{U}$. It is challenging to derive a tighter model confidence set with a closer-to-level- α coverage in cases when the distribution of $\mathbf{U}$ is unknown, since we often need the knowledge of $\mathbf{U}$'s distribution to obtain $B_\alpha(\tau, \mathbf{w})$ in (15) to quantify the uncertainty of $\mathbf{u}^{rel}$. This is because both τ and $\hat{\tau}(\mathbf{Y})$ are discrete, and typical concentration inequalities or central limit theorem do not apply. When the distribution of $\mathbf{U}$ is known but non-Gaussian, the approach in Algorithm 2 is not valid anymore because the independence among the three components in the decomposition (17) does not hold. In*

Algorithm 2 Confidence set construction for τ_0 **Input:** Design matrix $\mathbf{X}$, response vector $\mathbf{y}_{obs}$, candidate set $S^{(d)}$, simulation size J **Output:** Confidence set of τ_0 **Step 1:** For each candidate model $\tau_b \in S^{(d)}$, calculate $\hat{\tau}_{jb}^*, j = 1, \dots, J$.**for** $\tau_b \in S^{(d)}$ and $j \in 1, \dots, J$ **do****Step 1(a):** Calculate $\mathbf{w}_{obs} = (\mathbf{a}_{obs}, b_{obs}) = (\mathbf{H}_{\tau_b} \mathbf{y}_{obs}, \|(\mathbf{I} - \mathbf{H}_{\tau_b}) \mathbf{y}_{obs}\|)$.**Step 1(b):** Generate $\mathbf{u}_j^* \sim N(0, \mathbf{I}_n)$, and compute

$$\mathbf{y}_{jb}^* = \mathbf{a}_{obs} + b_{obs} \frac{(\mathbf{I} - \mathbf{H}_{\tau_b}) \mathbf{u}_j^*}{\|(\mathbf{I} - \mathbf{H}_{\tau_b}) \mathbf{u}_j^*\|}.$$

In addition, obtain the estimated model $\hat{\tau}_{jb}^* = \hat{\tau}(\mathbf{y}_{jb}^*)$ by

$$(22) \quad \hat{\tau}(\mathbf{y}_{jb}^*) = \arg \min_{\tilde{\tau} \in \mathcal{M}, \beta_{\tilde{\tau}} \in R^{|\tilde{\tau}|}} \|\mathbf{y}_{jb}^* - X_{\tilde{\tau}} \beta_{\tilde{\tau}}\|^2 \text{ s.t. } |\tilde{\tau}| \leq |\tau_b|.$$

end for**Step 2:** Estimate $p_{(\mathbf{w}_{obs}, \tau_b)}(\tilde{\tau})$ for all $\tilde{\tau} \in \mathcal{M}$ by $\hat{p}_{(\mathbf{w}_{obs}, \tau_b)}(\tilde{\tau}) = \frac{1}{J} \sum_{j=1}^J \mathbf{1}_{\{\hat{\tau}_{jb}^* = \tilde{\tau}\}}$, where $\mathbf{1}_{\{\cdot\}}$ is the indicator function.**Step 3:** Calculate $\tilde{T}(\mathbf{y}_{obs}, \tau_b) = \hat{\tau}_b^{obs}$ by $\hat{\tau}_b^{obs} = \arg \min_{\tilde{\tau} \in \mathcal{M}, \beta_{\tilde{\tau}} \in R^{|\tilde{\tau}|}} \|\mathbf{y}_{obs} - X_{\tilde{\tau}} \beta_{\tilde{\tau}}\|^2 \text{ s.t. } |\tilde{\tau}| \leq |\tau_b|$.**Step 4:** We then compute the estimated tail probability of $\tilde{T}(\mathbf{y}_{obs}, \tau_b)$ as

$$\hat{F}_{(\mathbf{w}_{obs}, \tau_b)}\left\{\tilde{T}(\mathbf{y}_{obs}, \tau_b)\right\} = \sum_{\{\tilde{\tau}: p_{(\mathbf{w}_{obs}, \tau_b)}(\tilde{\tau}) \leq p_{(\mathbf{w}_{obs}, \tau_b)}(\hat{\tau}_b^{obs})\}} \hat{p}_{(\mathbf{w}_{obs}, \tau_b)}(\tilde{\tau}).$$

We therefore obtain the level- α confidence set for τ_0

$$\bar{\Gamma}_{\alpha}^{\tau}(\mathbf{y}_{obs}) = \left\{ \tau_b \in S^{(d)} : \hat{F}_{(\mathbf{w}_{obs}, \tau_b)}\left\{\tilde{T}(\mathbf{y}_{obs}, \tau_b)\right\} \geq 1 - \alpha \right\}.$$

such cases, a profiling approach as described in [27, 59] can be potentially applied instead, although it may pose computational challenges and yield less efficient results. Alternatively, since we now have a model candidate set with a high coverage probability, we may follow [33] to construct a distribution-free confidence set for the best working models, that is, the model that provides the best predictive performance. The idea is to construct a confidence interval for the smallest cross-validation error, and use that to construct the confidence set for the best predictive model. Since the candidate models are low-dimensional and covers the truth τ_0 with a high probability, we believe that τ_0 is very likely to be the best predictive model among all the candidates. The theoretical justification and the coverage of this confidence set constructed using the “best predictive model” is an intriguing future research topic.

4. Inference for regression coefficients accounting for model selection uncertainty

Section 4.1 proposes a confidence set for any subset of $\beta_0^{full} = (\beta_{0,1}, \dots, \beta_{0,p})^\top$ that accounts for model selection uncertainty, and extend the work to make inference for any linear transformation of β_0^{full} . Section 4.2 discusses two special cases of practical importance: (a) inference for a single regression coefficient $\beta_{0,i}, i = 1, \dots, p$, and (b) joint inference for all regression coefficients β_0^{full} . Note that, most existing methods focus only on one of the two special cases, and there are few effective approaches on making inference for any subset or linear transformation of β_0^{full} in the literature. Moreover, our work guarantees both finite-sample and large-sample coverage, while existing methods provide only asymptotic inferences. In Section 4.1–4.2, we assume that the error terms are Gaussian $\mathbf{U} \sim N(0, \mathbf{I}_n)$. In Section 4.3, we extend the confidence sets for the regression coefficients in Section 4.1 to cases where non-Gaussian errors are present. Due to space limits, we defer the joint inference for the model and regression coefficients to Appendix B in [54].

4.1. *Inference for a subset of regression coefficients* Let $\beta_{0,\Lambda}$ be a collection of $\beta_{0,i}$ ’s that are of interests, where the index set $\Lambda \subset [p]$. The remaining parameters $\beta_{0,i}, i \notin \Lambda, \sigma_0$ and

τ_0 are nuisance parameters. The subset Λ is given based on the problem of interest, and it may overlap with or separate from τ_0 . Here our strategy is to first remove the influence of the nuisance parameters $\beta_{0,i}, i \notin \Lambda$ and σ_0 by defining a nuclear mapping function that only involves $\eta_\Lambda = (\beta_\Lambda, \tau)$, where τ is a potential value of τ_0 . The role of the nuclear mapping is similar to test statistics in the classical hypothesis testing framework, but in general, the definition of the nuclear mapping is broader and more flexible than the definition of test statistics. See [59] for a detailed discussion. We then utilize the model candidate set $S^{(d)}$ constructed in Section 2 to handle the impact of τ , leading to a valid confidence set for $\beta_{0,\Lambda}$.

For a given $\eta_\Lambda = (\beta_\Lambda, \tau)$, we define the nuclear mapping as follows

$$(23) \quad T(\mathbf{u}, \eta_\Lambda) = \begin{cases} \frac{\mathbf{u}^\top \mathbf{O}_{\tau,\Lambda} \mathbf{u}}{\mathbf{u}^\top (\mathbf{I} - \mathbf{H}_\tau) \mathbf{u} / (n - |\tau|)} & \text{if } \Lambda \cap \tau \neq \emptyset \text{ \& } \beta_i = 0 \text{ for any } i \in \Lambda \setminus \tau \\ \infty & \text{if } \beta_i \neq 0 \text{ for any } i \in \Lambda \setminus \tau \\ 0 & \text{if } \Lambda \cap \tau = \emptyset \text{ \& } \beta_i = 0 \text{ for any } i \in \Lambda \end{cases},$$

where $\mathbf{O}_{\tau,\Lambda}$ is the projection matrix of $(\mathbf{I} - \mathbf{H}_{\tau \setminus \Lambda}) \mathbf{X}_{\Lambda \cap \tau}$, and $\mathbf{H}_{\tau \setminus \Lambda}$ is the projection matrix of $\mathbf{X}_{\tau \setminus \Lambda}$. We can rewrite the above nuclear mapping as a function of $\mathbf{y}_\theta = \mathbf{X}_\tau \beta_\tau + \sigma \mathbf{u}$:

$$(24) \quad \tilde{T}(\mathbf{y}_\theta, \eta_\Lambda) = \begin{cases} \frac{(\mathbf{y}_\theta - \mathbf{X}_\Lambda \beta_\Lambda)^\top \mathbf{O}_{\tau,\Lambda} (\mathbf{y}_\theta - \mathbf{X}_\Lambda \beta_\Lambda)}{(\mathbf{y}_\theta - \mathbf{X}_\Lambda \beta_\Lambda)^\top (\mathbf{I} - \mathbf{H}_\tau) (\mathbf{y}_\theta - \mathbf{X}_\Lambda \beta_\Lambda) / (n - |\tau|)} & \text{if } \Lambda \cap \tau \neq \emptyset \text{ \& } \beta_i = 0 \text{ for any } i \in \Lambda \setminus \tau \\ \infty & \text{if } \beta_i \neq 0 \text{ for any } i \in \Lambda \setminus \tau \\ 0 & \text{if } \Lambda \cap \tau = \emptyset \text{ \& } \beta_i = 0 \text{ for any } i \in \Lambda \end{cases}.$$

Since when $\Lambda \cap \tau \neq \emptyset$ and $\beta_i = 0$ for any $i \in \Lambda \setminus \tau$, the distribution of the nuclear mapping defined above is $T(\mathbf{U}, \eta_\Lambda) = \tilde{T}(\mathbf{Y}_\theta, \eta_\Lambda) \sim F_{|\tau \cap \Lambda|, n - |\tau \cap \Lambda|}$, we let the Borel set be $B_{\eta_\Lambda}(\alpha) = [0, F_{|\tau \cap \Lambda|, n - |\tau \cap \Lambda|}^{-1}(\alpha)]$, such that $\mathbb{P}(\tilde{T}(\mathbf{Y}_\theta, \eta_\Lambda) \in B_{\eta_\Lambda}(\alpha)) \geq \alpha$. We can show that a valid level- α repro samples confidence set for $\eta_{0,\Lambda} = (\beta_{0,\Lambda}, \tau_0)$ is

$$(25) \quad \Gamma_\alpha^{\eta_\Lambda}(\mathbf{y}_{obs}) = \left\{ \eta_\Lambda : \tilde{T}(\mathbf{y}_{obs}, \eta_\Lambda) \in B_\alpha(\eta_\Lambda) \right\}.$$

Now with both $\beta_{0,i}, i \notin \Lambda$ and σ_0 out of the picture, we need to deal with the only remaining nuisance parameter τ . To handle the impact of τ , we utilize the model candidate set $S^{(d)}$ constructed in Section 2.2 and take a union approach. That is, for certain β_Λ , if (β_Λ, τ) is defined above for any τ in the candidate set $S^{(d)}$, we then retain the β_Λ in the confidence set for $\beta_{0,\Lambda}$. Specifically,

$$(26) \quad \begin{aligned} \Gamma_\alpha^{\beta_\Lambda}(\mathbf{y}_{obs}) &= \left\{ \beta_\Lambda : \tilde{T}(\mathbf{y}_{obs}, \eta_\Lambda) \in B_\alpha(\eta_\Lambda), \eta_\Lambda = (\beta_\Lambda, \tau) \text{ for some } \tau \in S^{(d)} \right\} \\ &= \bigcup_{\tau \in S^{(d)}} \left\{ \beta_\Lambda : \tilde{T}(\mathbf{y}_{obs}, \eta_\Lambda) \in B_\alpha(\eta_\Lambda), \eta_\Lambda = (\beta_\Lambda, \tau) \right\}. \end{aligned}$$

Note that this confidence set is a union of multiple smaller sets. This is different than confidence sets produced by existing methods which are often single intervals or ellipsoid sets. An illustration of such a confidence set is provided in the special cases that $\Lambda = [p]$, please see Figure A1 of Appendix F in [54].

We observe that inside the union in (26), each set is a confidence set based on certain low-dimensional model τ in the candidate set $S^{(d)}$. Although we do not know the true underlying model τ_0 , with Algorithm 1, we are able to construct a candidate set of reasonable size that would include τ_0 with a high probability. This enables us to guarantee the coverage rate, both in finite samples and asymptotically, as indicated in the following theorems.

THEOREM 10. *Under the conditions in Theorem 1, for any finite n and p and an arbitrarily small $\delta > 0$, the coverage probability of the confidence interval $\Gamma_\alpha^{\beta_\Lambda}(\mathbf{Y})$ defined in (26)*

371 is $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}\{\beta_{0,\Lambda} \in \Gamma_{\alpha}^{\beta_{\Lambda}}(\mathbf{Y})\} \geq \alpha - \delta - o(e^{-c_1 d})$ for some $c_1 > 0$. Further $\mathbb{P}_{\mathbf{Y}|\mathcal{U}^d}\{\beta_{0,\Lambda} \in$
 372 $\Gamma_{\alpha}^{\beta_{\Lambda}}(\mathbf{Y})\} \geq \alpha - \delta - o_p(e^{-c_1 d})$.

373 **THEOREM 11.** *Under the conditions in Theorem 2, for any finite d , the coverage probability*
 374 *of $\Gamma_{\alpha}^{\beta_{\Lambda}}(\mathbf{Y})$ defined in (26) is $\mathbb{P}_{(\mathcal{U}^d, \mathbf{Y})}\{\beta_{0,\Lambda} \in \Gamma_{\alpha}^{\beta_{\Lambda}}(\mathbf{Y})\} \geq \alpha - o(e^{-c_2 n})$ for some $c_2 > 0$.*
 375 *Further $\mathbb{P}_{\mathbf{Y}|\mathcal{U}^d}\{\beta_{0,\Lambda} \in \Gamma_{\alpha}^{\beta_{\Lambda}}(\mathbf{Y})\} \geq \alpha - o_p(e^{-c_2 n})$.*

REMARK 4 (Extension to make inference for any linear transformation of β_0^{full}). Let $\mathbf{L}\beta_0^{full}$ be a linear transformation of β_0^{full} , where $\mathbf{L}$ is a $l \times p$ transformation matrix. Let

$$\tilde{\mathbf{L}} = \begin{bmatrix} \mathbf{L} \\ \mathbf{I}_{(p-l) \times (p-l)}, \mathbf{0}_{(p-l) \times l} \end{bmatrix},$$

376 $\tilde{\beta}_0^{full} = \tilde{\mathbf{L}}\beta_0^{full}$, and $\tilde{\mathbf{X}} = \mathbf{X}\tilde{\mathbf{L}}^{-1}$. The inference for $\mathbf{L}\beta_0^{full}$ based on the data $(\mathbf{y}_{obs}, \mathbf{X})$ is now
 377 equivalently transformed to the inference for a subset of $\tilde{\beta}_0^{full}$ based on the transformed data
 378 $(\mathbf{y}_{obs}, \tilde{\mathbf{X}})$. Therefore we are able to construct the confidence set for $\mathbf{L}\beta_0^{full}$ by applying (26)
 379 on $(\mathbf{y}_{obs}, \tilde{\mathbf{X}})$. Note that one should also derive the candidate set $S^{(d)}$ from the transformed
 380 data $(\mathbf{y}_{obs}, \tilde{\mathbf{X}})$ using Algorithm 1.

381 **4.2. Two special cases of interest** As stated in [30, 61], we are often interested in marginal
 382 inference for a single regression coefficient $\beta_{0,i}$ in practice. Another interesting inference
 383 problem that has been studied in the literature is to jointly infer all regression coefficients
 384 β_0^{full} [17, 62]. In this subsection, we consider these interesting special cases and provide a
 385 new solution using the repro samples method.

Inference for a single regression coefficient. To obtain the repro samples confidence set for $\beta_{0,i}$, we simplify the nuclear mapping function defined in (23) and (24) by making $\Lambda = \{i\}$. For a given $\eta_i = (\beta_i, \tau)$, it is

$$T(\mathbf{u}, \eta_i) = \tilde{T}(\mathbf{y}_{\theta}, \eta_i) = \begin{cases} \frac{\mathbf{u}^{\top} \mathbf{O}_{\tau, i} \mathbf{u}}{\mathbf{u}^{\top} (\mathbf{I} - \mathbf{H}_{\tau}) \mathbf{u} / (n - |\tau|)} = \frac{(\mathbf{y}_{\theta} - \mathbf{X}_i \beta_i)^{\top} \mathbf{O}_{\tau, i} (\mathbf{y}_{\theta} - \mathbf{X}_i \beta_i)}{(\mathbf{y}_{\theta} - \mathbf{X}_i \beta_i)^{\top} (\mathbf{I} - \mathbf{H}_{\tau}) (\mathbf{y}_{\theta} - \mathbf{X}_i \beta_i) / (n - |\tau|)} & \text{if } i \in \tau \\ \infty & \text{if } i \notin \tau, \beta_i \neq 0, \\ 0 & \text{if } i \notin \tau, \beta_i = 0 \end{cases}$$

386 where $\mathbf{O}_{\tau, i}$ is the projection matrix of $(\mathbf{I} - \mathbf{H}_{\tau_{-i}}) \mathbf{X}_i$, and $\tau_{-i} = \tau \setminus \{i\}$. Note that for $i \in \tau$, the
 387 nuclear statistics $\tilde{T}(\mathbf{y}_{\theta}, \eta_i)$ is equivalent to the square of t -statistics for testing $H_0 : \beta_{0,i} = \beta_i$,
 388 obtained by fitting a linear regression of $\mathbf{y}_{\theta}$ on $\mathbf{X}_{\tau}$.

Then following (26), we obtain the confidence set for $\beta_{0,i}$,

$$\begin{aligned} \Gamma_{\alpha}^{\beta_i}(\mathbf{y}_{obs}) &= \left\{ \beta_i : \tilde{T}(\mathbf{y}_{obs}, \eta_i) \in B_{\alpha}(\eta_i), \eta_i = (\beta_i, \tau) \text{ for some } \tau \in S^{(d)} \right\} \\ (27) \quad &= \bigcup_{\tau \in S^{(d)}} \left\{ \beta_i : \tilde{T}(\mathbf{y}_{obs}, (\beta_i, \tau)) \leq F_{1, n-|\tau|}^{-1}(\alpha) \right\}, \end{aligned}$$

389 where we let $B_{\alpha}(\eta_i) = B_{\alpha}(\tau) = [0, F_{1, n-|\tau|}^{-1}(\alpha)]$. Then by Theorems 10 and 11, $\Gamma_{\alpha}^{\beta_i}(\mathbf{y}_{obs})$ is
 390 a level- α confidence set for $\beta_{0,i}$.

391 **REMARK 5** (Comparison with the debiased method). *We discuss the difference between*
 392 *our method and the debiased Lasso. First of all, our method offers the finite-sample coverage*
 393 *guarantee, while the debiased Lasso method can only achieve the asymptotic coverage rate.*
 394 *More specifically, the debiased Lasso method needs the sample size $n \rightarrow \infty$ to make sure the*
 395 *bias, which comes from the regularized estimation and is of order $O(|\tau_0| \sqrt{\log p/n})$, goes to 0.*
 396 *In contrast, our method bypasses the estimation step and constructs the confidence sets directly*
 397 *via the repro sampling framework, and is therefore unbiased in nature. Second, the debiased*

Lasso method [30] is designed to make inferences for an individual regression coefficient. The idea of the debiased Lasso method was later generalized to make inferences for functions of the regression coefficients, such as co-heritability[23] and group inference statistics [22]. However, such a generalization relies on specific forms of the functions and does not allow arbitrary functions. Our method, however, as we will show in Remark 6, can be used to construct the confidence sets for arbitrary functions. Third, we will show in Section 5.2 that, when the sample size is small, the debiased Lasso method may have either coverage issues or overly large intervals for large regression coefficients. In contrast, our method achieves the nominal coverage for both large and zero regression coefficients in the small sample setting with preferable interval lengths. See Section 5.2 for more details.

Joint inference for all regression coefficients. Let $\Lambda = [p]$ and we make joint inference for all regression coefficients β_0^{full} here. Note that $\tau \subset \Lambda$, so $\tau \setminus \Lambda = \emptyset$. Following (23) and (24), the nuclear mapping function for $\eta = (\beta, \tau) = ((\beta_\tau, \mathbf{0}_{\tau^c}), \tau)$ is

$$\begin{aligned} T(\mathbf{u}, \eta) &= \frac{\mathbf{u}^\top \mathbf{H}_\tau \mathbf{u} / |\tau|}{\mathbf{u}^\top (I - \mathbf{H}_\tau) \mathbf{u} / (n - |\tau|)} \\ (28) \quad &= \frac{(\mathbf{y}_\theta - \mathbf{X}_\tau \beta_\tau)^\top \mathbf{H}_\tau (\mathbf{y}_\theta - \mathbf{X}_\tau \beta_\tau) / |\tau|}{(\mathbf{y}_\theta - \mathbf{X}_\tau \beta_\tau)^\top (I - \mathbf{H}_\tau) (\mathbf{y}_\theta - \mathbf{X}_\tau \beta_\tau) / (n - |\tau|)} := \tilde{T}(\mathbf{y}_\theta, \eta). \end{aligned}$$

We then let the Borel set be $B_\alpha(\eta) = B_\alpha(\tau) = [0, F_{|\tau|, n-|\tau|}^{-1}(\alpha)]$, and construct the joint confidence set for β_0^{full} following from (26),

$$\begin{aligned} \Gamma_\alpha^\beta(\mathbf{y}_{obs}) &= \left\{ \beta : \tilde{T}(\mathbf{y}_{obs}, \eta) \leq F_{|\tau|, n-|\tau|}^{-1}(\alpha), \beta = (\beta_\tau, \mathbf{0}_{\tau^c}), \eta = (\beta, \tau) \text{ for some } \tau \in S^{(d)} \right\} \\ (29) \quad &= \bigcup_{\tau \in S^{(d)}} \left\{ \beta : \tilde{T}(\mathbf{y}_{obs}, \eta) \leq F_{|\tau|, n-|\tau|}^{-1}(\alpha), \beta = (\beta_\tau, \mathbf{0}_{\tau^c}), \eta = (\beta, \tau) \right\}. \end{aligned}$$

Again, following Theorems 10 and 11, $\Gamma_\alpha^\beta(\mathbf{y}_{obs})$ is a level- α confidence set for the entire coefficients vector β_0^{full} . In addition, to better understand the confidence set in (29), we have presented a visualization of (29) for an example in Appendix F in [54].

REMARK 6 (Extension to inference for any function of β_0^{full}). We can extend the joint confidence set in (29) to obtain a repro samples confidence set for any function of β_0^{full} , say $h(\beta_0^{full})$. To put it more clearly, for each (τ, β_τ) in the confidence set (29), we collect the function value $h(\beta_\tau^{full})$, where $\beta_\tau^{full} = (\beta_\tau, \mathbf{0}_{\tau^c})$ to form the confidence set for $h(\beta_0^{full})$, i.e. $\Gamma_\alpha^h(\mathbf{y}_{obs}) = \{h((\beta_\tau, \mathbf{0}_{\tau^c})) : \tilde{T}(\mathbf{y}_{obs}, \eta) \leq F_{|\tau|, n-|\tau|}^{-1}(\alpha), \eta = (\tau, \beta_\tau) \text{ for } \tau \in S^{(d)}\}$.

4.3. Extension to models with non-Gaussian errors In this section, we extend the confidence set for the regression coefficients in (26) to models with non-Gaussian errors. We first introduce the following corollary.

COROLLARY 1. Let $\tilde{\Gamma}^{\beta_\Lambda|\tau}(\mathbf{y}_{obs})$ be a data-dependent set of β_Λ that also depends on a given model τ , and let $\tilde{\alpha} = \mathbb{P}(\beta_{0,\Lambda} \in \tilde{\Gamma}^{\beta_\Lambda|\tau_0}(\mathbf{Y}))$ be the coverage probability for $\beta_{0,\Lambda}$ given the true model τ_0 , then the confidence set for $\beta_{0,\Lambda}$ constructed by $\tilde{\Gamma}^{\beta_\Lambda}(\mathbf{y}_{obs}) = \bigcup_{\tau \in S^{(d)}} \tilde{\Gamma}^{\beta_\Lambda|\tau}(\mathbf{y}_{obs})$ has a coverage rate bounded by $\mathbb{P}(\beta_{0,\Lambda} \in \tilde{\Gamma}^{\beta_\Lambda}(\mathbf{Y})) \geq \tilde{\alpha} - \mathbb{P}(\tau_0 \notin S^{(d)})$.

The above corollary implies that models with non-Gaussian errors can still follow the idea in (26) to construct confidence sets for the regression coefficients. This is because the model candidate set proposed in Section 2.2 still covers the true model τ_0 with a high probability according to the results in Section 2.4 for common non-Gaussian errors. Once we have a candidate set $S^{(d)}$ that guarantees to cover the true model such that $\mathbb{P}(\tau_0 \notin S^{(d)})$ is close to 0, all we need is a valid inference approach in low-dimensional settings that can achieve

the desired coverage given τ_0 . In other words, we only need a data-dependent set given a low-dimensional model τ , namely $\tilde{\Gamma}^{\beta_\Lambda|\tau}(\mathbf{y}_{obs})$, such that given the true model, $\tilde{\Gamma}^{\beta_\Lambda|\tau_0}(\mathbf{y}_{obs})$ is a valid confidence set for $\beta_{0,\Lambda}$ with a desired coverage rate of (approximately) α . It then follows immediately from Corollary 1 that $\tilde{\Gamma}^{\beta_\Lambda}(\mathbf{y}_{obs}) = \bigcup_{\tau \in S^{(d)}} \tilde{\Gamma}^{\beta_\Lambda|\tau}(\mathbf{y}_{obs})$ is a valid level- α confidence set for $\beta_{0,\Lambda}$.

Fortunately, such a $\Gamma^{\beta_\Lambda|\tau}(\mathbf{y}_{obs})$ is not difficult to find, since the inference for linear models with non-Gaussian errors in low-dimensional settings have been extensively studied. For example, [32] proposed a likelihood-based approach for t and *Cauchy* errors, and [43] provided a thorough review on different approaches to deal with non-Gaussianity in the error terms of linear models. To make inference for a model with contaminated errors, see [14, 35] and the reference therein. For sub-Gaussian errors, it is well known that most of the results for Gaussian errors hold asymptotically due to the central limit theorem [58].

5. Simulation studies In this section, we conduct simulation studies to evaluate the numerical performance of the proposed repro samples methods. The synthetic data are generated from the following five models:

(M1) (Extremely high dimension) Let $\beta_0^{full} = (3, 2, 1.5, 0, \dots, 0)$. For $j_1, j_2 \in [p]$, the correlation between x_{j_1} and x_{j_2} is set to $0.5^{|j_1-j_2|}$. We set $n = 50, p = 1000$ and $\sigma = 1$.

(M2) (Decaying signal) Let $\beta_0^{full} = (2, 1.5, 1, 0.8, 0.6, 0, \dots, 0)$. For $j_1, j_2 \in [p]$, the correlation between x_{j_1} and x_{j_2} is set to $0.1^{|j_1-j_2|}$. We let $n = 80, p = 150$ and $\sigma = 1$.

(M3) (High-dimensional, decaying signal) Let $\beta_0^{full} = (3, 2, 1.5, 1, 0.8, 0.6, 0, \dots, 0)$. The correlation between x_{j_1} and x_{j_2} is $0.1^{|j_1-j_2|}$, $j_1, j_2 \in [p]$. Let $n = 100, p = 500$, and $\sigma = 1$.

(M4) (From [36] with many signals, but changed n to create a high-dimensional setting with $p > n$) Let the true values $\beta_0^{full} = (1, \dots, 1, 0, \dots, 0)$, where the first 12 coefficients $\beta_{0,j} = 1$, for $1 \leq j \leq 12$, and remaining $\beta_{0,j} = 0$, for $j \geq 12$. For $j_1, j_2 \in [p]$, the correlation between x_{j_1} and x_{j_2} is set to $0.5^{|j_1-j_2|}$. We let $n = 150, p = 200$ and $\sigma = 1$.

(M5) (Perfect colinearity, restricted eigenvalue conditions do not hold) Let the true values $\beta_0^{full} = (3, 2, 1.5, 0, \dots, 0)$. For $j_1, j_2 \in [p]$, the correlation between x_{j_1} and x_{j_2} is set to $0.5^{|j_1-j_2|}$. To create perfect collinearity, we set $2x_{10} + 3x_{11} + x_{10} = 3x_1 + 2x_2 + 1.5x_3$, such that the restricted eigenvalue condition does not hold, and $C_{\min} = 0$. We set $n = 50, p = 1000$ and $\sigma = 1$.

The first model (M1) represents an extremely high-dimensional setting with $p \gg n$. The second model (M2) represents a challenging case of decaying signals with the weakest one just 0.6. We set (M3) by increasing the dimension of (M2) to study the performance of the proposed approach when both a high-dimensional design matrix and weak signals are present in the data. We also add a strong signal of $\beta_{1,0} = 3$ so the range of signal strength is from 0.6 to 3. Model (M4), a high-dimensional adaptation of the scenario (f) in [36], features a setting that involves many true signals. Finally, we present an extremely challenging case in (M5), where typical conditions in the literature such as the restricted eigenvalue condition do not hold anymore. We replicate the simulation for 200 times for each model.

5.1. Model candidates and inference for the true model τ_0 We first study the numerical performance of the data-driven model candidate set $S^{(d)}$ in (6), produced by Algorithm 1, and the 95% repro samples model confidence set in (21), constructed by Algorithm 2.

To motivate the use of a set of models to quantify model selection uncertainty, we first examine the performance of a single selected (estimated) model based on common model selection criteria, AIC, BIC, cross-validation (CV), and the knockoff [7]. Table 1 reports the proportions of times a single selected model correctly identifies the true model under (M1)–(M5). We observe that the single selected model frequently differs from the true model:

TABLE 1
Percentage of times when point estimation is the true model among the 200 simulations.

Model	AIC	BIC	CV	Knockoff
M1	0.695	0.790	0.645	0.000
M2	0.250	0.510	0.205	0.000
M3	0.780	0.810	0.750	0.000
M4	0.730	0.860	0.655	0.600
M5	0.845	0.880	0.830	0.000

for Model (M2), the majority of the times the estimated model is wrong; for the other three settings, AIC, BIC, and CV methods are correct 60%–80% of the times. The knockoff approach performs poorly in estimating the true model, since it is developed mainly to control the false discovery rate. Moreover, there is no finite-sample theory to ensure that the estimated models are accurate, and even asymptotic results depend on assumptions that are often hard to verify. Therefore, incorporating data uncertainty into model selection or estimation is essential, underscoring the importance of using a valid and efficient model confidence set, such as the one we have proposed.

In our implementation of Algorithm 1, we use the EBIC [10] to select the tuning parameter when constructing the model candidate set. Further implementation details are provided in Appendix F in [54]. When applying Algorithm 2 to obtain our model confidence set, we calculate the $\hat{\tau}(\mathbf{y}_j^*)$ in (22) by obtaining the largest estimated model that is not larger than $|\tau_b|$ in the adaptive Lasso solution path. Also, in our analysis, we set the number of repro samples for the candidate set in Algorithm 1 to be $d = 1000$ for Models (M1), (M4), and (M5). For Models (M2)–(M3) with weak signals, identifying the true model is a known challenging problem. In this case, we set the number of repro samples to be a large $d = 10,000$ for (M2) and $d = 100,000$ for (M3). Regarding the number of repro samples in Algorithm 2 for calculating the distribution of the nuclear statistics, we set $J = 200$ for all five models.

We compare our repro samples approach with the residual bootstrap approach in the literature (e.g., [9]). The numbers of bootstrap samples are 1000 for (M1), (M4) and (M5), and 10,000, and 100,000 for (M2) and (M3) respectively, matching the numbers of repro samples for used for finding the candidate models. In each setting, the collection of all models obtained using the bootstrap samples forms a bootstrap model candidate set. Here, to implement the bootstrap approach, we use AIC, BIC, and CV to choose the tuning parameter. The bootstrap model “confidence” sets are obtained by removing the least frequent model estimations from the bootstrap candidate model set, with the total (cumulative) frequency of the removed models not larger than 5%. We note that the bootstrap method here is an ad hoc method commonly used in current practice. Due to the discreteness of the model space and estimated models, there is no theoretical support for the “confidence” claim that such a bootstrap method can get a valid level-95% model confidence set for the true model τ_0 .

Table 2, columns 3–4, compares the model candidate sets from the proposed repro samples approach and the residual bootstrap approaches with different tuning criteria. We report the average cardinality of the model candidate sets (Cardinality) and the percentage of simulation cases where the true model τ_0 is included (Inclusion of τ_0). From Table 2, we see that the proposed repro samples approach provides much smaller model candidate sets. For (M1) where $n = 50, p = 1000$, the repro samples candidate sets achieve 100% coverage for τ_0 with a size of only 2–3 on average out of the 1000 repro samples. Even for (M2) and (M3), where weak signals make identifying the true model notoriously challenging, the proposed procedure produces a candidate set of around 30 and 5 models on average, respectively, while covering the true model 98% and 99.5% of the times. The bootstrap method, on the other hand, yields 50–600 models out of 1000 replicates for (M1), (M4) and (M5), around 1700–4000 models out of 10,000 bootstrap runs for (M2), and around 3000–6500 models out of 100,000

TABLE 2
Performances of Model Candidate Sets and 95% Model Confidence Sets for the True Model τ_0

Model	Method	Model Candidate Sets		Level-95% Model Confidence Sets	
		Cardinality of $S^{(d)}$	Inclusion of τ_0	Cardinality of $\Gamma_{0.95}^\tau$	Coverage of $\Gamma_{0.95}^\tau$
M1	Repro samples	2.605 (0.191)	1.000 (0.000)	2.180 (0.102)	1.000 (0.000)
	Bootstrap AIC	215.425 (10.855)	1.000 (0.000)	165.960 (7.201)	1.000 (0.000)
	Bootstrap BIC	146.100 (7.423)	1.000 (0.000)	110.440 (5.423)	1.000 (0.000)
	Bootstrap CV	259.535 (11.891)	1.000 (0.000)	207.500 (7.890)	1.000 (0.000)
M2	Repro samples	29.455 (3.080)	0.980 (0.010)	12.050 (0.708)	0.955 (0.015)
	Bootstrap AIC	4350.850 (134.000)	1.000 (0.000)	4350.850 (134.000)	0.995 (0.005)
	Bootstrap BIC	2303.190 (75.708)	1.000 (0.000)	2286.015 (77.898)	0.995 (0.005)
	Bootstrap CV	5033.700 (134.233)	1.000 (0.000)	5033.700 (134.233)	0.995 (0.005)
M3	Repro samples	4.710 (0.558)	0.995 (0.005)	3.810 (0.316)	0.985 (0.009)
	Bootstrap AIC	5088.030 (456.021)	1.000 (0.000)	3481.215 (497.552)	0.935 (0.017)
	Bootstrap BIC	2944.325 (245.670)	1.000 (0.000)	1425.125 (265.383)	0.905 (0.021)
	Bootstrap CV	6458.345 (570.104)	1.000 (0.000)	4989.195 (618.298)	0.955 (0.015)
M4	Repro samples	1.875 (0.144)	0.995 (0.005)	1.745 (0.087)	0.995 (0.005)
	Bootstrap AIC	165.410 (6.178)	1.000 (0.000)	162.250 (6.495)	1.000 (0.000)
	Bootstrap BIC	96.240 (3.210)	1.000 (0.000)	85.415 (4.107)	1.000 (0.000)
	Bootstrap CV	204.175 (7.291)	1.000 (0.000)	203.090 (7.414)	1.000 (0.000)
M5	Repro samples	2.775 (0.199)	1.000 (0.000)	2.640 (0.164)	0.985 (0.009)
	Bootstrap AIC	72.240 (6.341)	0.990 (0.007)	57.875 (6.820)	0.990 (0.007)
	Bootstrap BIC	55.285 (4.794)	0.990 (0.007)	41.410 (5.191)	0.990 (0.007)
	Bootstrap CV	87.515 (7.406)	0.990 (0.007)	74.065 (7.925)	0.990 (0.007)

bootstrap samples for (M3), proportionate to a random search. In summary, Table 2 clearly demonstrates the advantage of our proposed method in finding candidate models. The size of the candidate model set by our repro samples method is small and manageable, while those by the corresponding bootstrap methods are all unreasonably large, making them inapplicable in practice. Furthermore, following a reviewer’s suggestion, we have also tried constructing an alternative model candidate set by including all models on the solution path of EBIC using just the original data without the repro samples procedure. The empirical coverage of such an EBIC candidate set is not satisfactory except for (M1), as the coverage for (M2)–(M5) ranges only from 81.5% to 94.5%, which is not comparable to our repro samples approach in general.

Table 2, columns 5–6, reports the average cardinality of the confidence sets obtained using Algorithm 2 and the bootstrap approaches, along with their coverage of the true model τ_0 out of the 200 repetitions. From Table 2, we see that, for (M1), (M4) and (M5), the model confidence set based on the repro samples approach only contains 1.7–2.6 models on average, while the “confidence” sets by the bootstrap methods have sizes between 40–600. For (M2) and (M3), the model confidence sets generated by the bootstrap are impractically large, containing between 1400–5000 models on average. Even with those many models, the bootstrap confidence sets with AIC and BIC slightly undercover the true model τ_0 for (M3). In contrast, for (M2) and (M3), the repro samples approach achieves much more efficient results: its confidence sets contain only about 12 and 4 models on average, respectively, yet still maintain coverage above 95%. This strong performance holds even under the challenging decaying-signal scenarios, where the smallest nonzero coefficient is as low as 0.6.

The empirical coverage rate of our model confidence set is higher than 95% except for (M2). This is mostly due to the discrete nature of the inference target here, the true model τ_0 . Unlike a typical confidence interval for a parameter whose value is a real number, the coverage rate for τ_0 is by no means a continuous or linear function of the size of the confidence set. Therefore, reducing the size of the confidence set will reduce the coverage rates below the desired level of 95%. However, there would be less overcoverage when the candidate set contains more models, as in (M2), where the coverage rate of the confidence set is 95.5%.

To sum up, the results in Table 2 demonstrate that the proposed repro samples method constructs valid and efficient model confidence sets for the true model τ_0 , even under

TABLE 3
Repro Confidence Sets for (Scalar) Regression Parameter $\beta_{0,i}$ with Comparison to Debiased Lasso

Model	$\beta_{0,i}$	Repro Samples		Debiased Lasso (JM)		Debiased Lasso (ZZ)	
		Coverage	Width	Coverage	Width	Coverage	Width
M1	All $\beta_{0,i}$	1.000(0.000)	0.003(0.000)	0.984(0.000)	0.247(0.000)	0.961(0.000)	1.335(0.000)
	$\beta_{0,i} \neq 0$	0.952(0.009)	0.748(0.006)	0.357(0.020)	0.247(0.005)	0.982(0.005)	1.321(0.009)
	$\beta_{0,i} = 0$	1.000(0.000)	0.001(0.000)	0.986(0.000)	0.247(0.000)	0.961(0.000)	1.335(0.000)
M2	All $\beta_{0,i}$	0.999(0.000)	0.038(0.000)	0.982(0.001)	0.498(0.000)	0.983(0.001)	0.942(0.001)
	$\beta_{0,i} \neq 0$	0.975(0.005)	0.539(0.003)	0.938(0.008)	0.497(0.002)	0.989(0.003)	0.939(0.004)
	$\beta_{0,i} = 0$	1.000(0.000)	0.021(0.000)	0.984(0.001)	0.498(0.000)	0.983(0.001)	0.942(0.001)
M3	All $\beta_{0,i}$	1.000(0.000)	0.007(0.000)	0.989(0.000)	0.429(0.000)	0.974(0.001)	0.943(0.000)
	$\beta_{0,i} \neq 0$	0.959(0.006)	0.459(0.002)	0.868(0.010)	0.428(0.001)	0.977(0.004)	0.943(0.004)
	$\beta_{0,i} = 0$	1.000(0.000)	0.002(0.000)	0.991(0.000)	0.429(0.000)	0.974(0.001)	0.943(0.000)
M4	All $\beta_{0,i}$	0.997(0.000)	0.027(0.001)	0.985(0.001)	0.441(0.000)	0.972(0.001)	0.513(0.000)
	$\beta_{0,i} \neq 0$	0.957(0.004)	0.437(0.001)	0.942(0.005)	0.439(0.001)	0.953(0.004)	0.511(0.001)
	$\beta_{0,i} = 0$	1.000(0.000)	0.001(0.000)	0.988(0.001)	0.441(0.000)	0.974(0.001)	0.513(0.000)
M5	All $\beta_{0,i}$	1.000(0.000)	0.003(0.000)	0.984(0.000)	0.244(0.000)	NA	NA
	$\beta_{0,i} \neq 0$	0.953(0.009)	0.755(0.006)	0.337(0.019)	0.246(0.005)	NA	NA
	$\beta_{0,i} = 0$	1.000(0.000)	0.001(0.000)	0.986(0.000)	0.244(0.000)	NA	NA

the challenging settings among (M1)–(M5). In contrast, the bootstrap method exaggerates the uncertainty of model selection by producing extremely large number of models in its “confidence” sets, rendering results that are not very useful in practice.

5.2. Inference for regression coefficients accounting for model selection uncertainty Here, we examine the performance of the proposed repro samples confidence set for a single regression coefficient in (27) and compare it with the state-of-the-art debiased methods. Due to space limits, the results of the joint confidence set for β_0^{full} in (29) are placed in Appendix F in [54].

Table 3 compares the proposed 95% repro samples confidence sets for single regression coefficients with the two state-of-the-art debiased approaches from [30] (debiased Lasso (JM), authors’ code at [29]) and [61] (debiased Lasso (ZZ), authors’ R code). Although the debiased Lasso (JM) can achieve the desired coverage rate when averaging over all β_j ’s, it significantly undercovers the non-zero coefficients (signals) for all models except for (M4), particularly for (M1) where the correlations among the covariates are large, and for (M5) where the restricted eigenvalue condition does not hold. This undercoverage, noted by both [61] and [30], arises because large correlations inflate the estimation error $|\hat{\beta}_i - \beta_{0,i}|$ for nonzero signals when sample sizes are limited. [61] provides an enhanced method (ZZ) to overcome this issue by including an independent set of highly correlated variables when debiasing to enforce small correlations between the score vector and covariates. This approach improves the coverage rates in the three simulation settings, but at the expense of larger interval widths. In contrast, the repro samples method consistently achieves the desired coverage for all the signals in all five models, with confidence sets at least 40% shorter than those from the debiased Lasso (ZZ). Besides, the debiased Lasso (ZZ) fails to produce results for (M5) because the required condition on the design matrix is not satisfied in this case. Additionally, we also compare the confidence intervals of each non-zero coefficients, confirming the aforementioned advantages of our repro samples method over the debiased approaches. See Appendix F in [54] for the details.

Moreover, compared to both debiased approaches, the repro samples confidence sets for zero coefficients appear to be much narrower, making the average width of the proposed confidence sets much smaller. In addition, the computing code of the enhanced debiased Lasso (ZZ) approach requires arbitrarily pre-selecting the number of highly correlated variables to calculate the score vector. Conversely, the repro samples approach is a data-dependent procedure that avoids such an ad hoc decision. In summary, the repro samples approach achieves better coverage with data-dependent smaller confidence set for a single regression coefficient.

TABLE 4
Repro Samples Candidate Models and Top 10 Variables from the Bootstrap

(a) <i>Repro samples candidate models</i>		(b) <i>Top 10 most often selected variables by the bootstrap</i>					
Candidate Model	Tail Probability	AIC		BIC		CV	
ϕ	1.000	ZNF208	4877	ZNF208	4350	ZNF208	4727
(ZNF208)	1.000	CCDC19	4751	CCDC19	2259	CCDC19	3946
(ZNF208, NETO2)	1.000	SAA4	4319	GRM8	2215	GRM8	3643
(ZNF208, TOP1)	0.265	GRM8	4233	TOP1	2192	SAA4	3517
(ZNF208, GRM8)	0.000	SLC25A23	3975	NETO2	1932	CETP	3436
(ZNF208, RPS28, TOP1)	0.000	ATP5G3	3903	CETP	1915	NETO2	3134
		NETO2	3820	SAA4	1901	TOP1	3036
		CLCNKB	3769	SLC25A23	1405	SLC25A23	2868
		CETP	3754	HCRTR2	1210	PAX3	2300
		CPNE1	3718	PAX3	979	ATP5G3	2182

6. Real Data Analysis In this section, we apply our repro samples method along with the bootstrap approaches to analyze the glioblastoma gene expression data from The Cancer Genome Atlas (TCGA) consortium. The goal of the analysis is to identify the highly informative genes regarding the survival time of glioblastoma. The gene expression data set we obtained, which includes 428 samples, were measured experimentally by the University of North Carolina TCGA genomic characterization center. In our analysis, the logarithm of the survival time is the response variable. We first screen the 17814 genes using sure independence screening [20] to identify 1000 genes that are most correlated with the response [55].

Here we set the number of the repro samples $d = 5000$ when constructing the model candidate set with Algorithm 1. Similar to the simulation studies, we compare the proposed repro samples procedure with the bootstrap approaches for inference on the true models. Out of the 5000 iterations, bootstrap yields 4935 candidate models with BIC, 5000 with AIC or CV. This means almost every bootstrap sample will generate a different model, and the candidate models exhaust all the 1000 covariates, confirming earlier findings in Section 5.1 that bootstrap includes too many models with spurious variables. In contrast, the repro samples method identifies six candidate models out of 5000 iterations, as shown in Table 4(a). The results are also consistent with the previous findings from the simulations, demonstrating the efficient construction of candidate sets by the repro samples method. Moreover, based on the six candidate models, we follow Algorithm 2 to construct a model confidence set with $J = 200$. We summarize the tail probability $\hat{F}_{(\mathbf{w}_{obs}, \tau_b)}\{\tilde{T}(\mathbf{y}_{obs}, \tau_b)\}$ in Algorithm 2, also interpreted as a p-value, in Table 4(a). According to the tail probabilities, the 95% model confidence set consists of four models: ϕ , (ZNF208), (ZNF208, TOP1), and (ZNF208, NETO2).

Comparing with the top covariates obtained using the bootstrap, listed in Table 4(b), the repro samples confidence set highlights two key genes: TOP1 and NETO2. The significance of TOP1 gene in glioblastoma diagnosis and treatment has been proven by numerous studies [e.g., 44, 56]. Furthermore, the link between the NETO2 gene and glioblastoma is well documented by a recent study in [37]. Meanwhile, CCDC19—the second most frequent gene under the bootstrap method—does not appear in our candidate models, and there is no established evidence supporting its association with glioblastoma. Overall, existing scientific findings suggest that the proposed repro samples procedure is more reliable when inferring the true underlying models.

7. Discussion We have developed a repro samples approach to address inference problems concerning high-dimensional linear models. The paper contains three technical innovations.

1. We develop a data-driven approach to obtain an efficient model candidate set, which covers the true model with high probability by including just a reasonable number of model candidates. Using this model candidate set effectively addresses the computational issue since it avoids searching the entire model space. The approach is based on the matching attempt of repro samples with the observed data, leading to the many-to-one mapping

function in (5). Specifically, this many-to-one mapping tells us that there always exists a neighborhood of $\mathbf{U}$, within which a repro copy $\mathbf{U}^*$ can help recover the true model with a high probability. With this insight, we propose a formal procedure and provide supporting theories and numerical evidence, both of which also help to outline trade-offs among sample size, the signal strength, and the performance of the model candidate set. Additionally, the number of repro samples d in Algorithm 1 is analog to the bootstrap sample size in the bootstrap approach, and our simulation study shows empirically that d does not need to be exceptionally large. In theory, we have derived a theorem, which is placed in Appendix E [54] with its discussions due to space constraints, to study the scale of d required to theoretically guarantee the model candidate set includes the true model τ_0 .

Moreover, we have developed supporting theoretical results that accommodate both Gaussian and common non-Gaussian error distributions. We further extend the finite-sample result for sub-Gaussian errors in Theorem 5 to any continuous error distribution with a finite second moment, as stated in the following corollary.

COROLLARY 2. *Suppose $\mathbf{Y}$ is generated by (2) with $\mathbf{U}$ being a continuous random vector with a finite second moment $\mathbb{E}(\|\mathbf{U}\|^2) < \infty$, then when $\lambda \in [n\gamma_\delta^{3/4}, n\gamma_\delta^{1/4} \frac{C_{\min}}{6}]$, the probability bound (10) in Theorem 5 still holds.*

In addition, our model candidate set can be used to achieve model selection consistency under non-Gaussian errors. Specifically, Section 4.2 of [33] shows that the smallest model in their cross-validation confidence set is consistent in model selection under the assumption of a finite sixth moment if the candidate models include the true model. By Corollary 2, the proposed model candidate set $S^{(d)}$ contains τ_0 with high probability for sufficiently large d under a finite second-moment assumption; hence, under the stronger sixth-moment condition required by [33], the inclusion still holds, and applying their result to our repro-samples candidate set $S^{(d)}$ delivers selection consistency. Alternatively, Theorem 6 implies the same consistency under sub-Gaussian errors for suitably small d .

2. When making inference for the true model τ_0 , we develop a conditional repro samples approach to remove the impact of the nuisance parameters (β_τ, σ^2) . This conditional approach works in general for inference problems beyond the scope of this paper. In particular, let $\theta_0 = (\nu_0, \xi_0)$, where ν_0 and ξ_0 are the target and nuisance parameters, respectively. If we have a nuclear mapping $T(\mathbf{U}, \theta)$ and a quantity $\mathbf{W}(\mathbf{U}, \theta)$, such that the conditional distribution of $T(\mathbf{U}, \theta)$ given $\mathbf{W}(\mathbf{U}, \theta) = \mathbf{w}$ is free of ξ_0 , then there exists a Borel set $B_\alpha(\nu, \mathbf{w})$ free of the nuisance ξ such that $\mathbb{P}(T(\mathbf{U}, \theta) \in B_\alpha(\nu, \mathbf{w}) | \mathbf{W}(\mathbf{U}, \theta) = \mathbf{w}) \geq \alpha$. Consequently, similar to (16), we obtain a valid level- α confidence set for ν_0 .
3. We propose confidence sets both for a single and for any subset of regression coefficients. In contrast, existing literature only focuses on one aspect of these inference problems. This is because, unlike existing approaches, we take a union of intervals or multi-dimensional ellipsoids based on each low-dimensional model in the model candidate set. Therefore, our approach takes into account the uncertainty in model estimation. Not only does it provide the desired coverage, it also produces confidence sets that are sparse and generally smaller than the existing methods, including the debiased approach. We also would like to note that the computational complexity of the proposed repro samples procedure is $O(dp^3)$, while for debiased Lasso it is $O(p^4)$. Therefore the proposed procedure is likely to be more efficient computationally in high-dimensional settings when $p \gg n$.

Finally, there are several potential directions for extensions of the work. First, it is possible to extend the proposed approach to a generalized linear model, where two challenges arise: (a) the geometry between the underlying linear function, error term and the response is much more complicated; (b) the conditional approach to handle nuisance regression parameters does not directly apply. We have reported on how to overcome these challenges for binary regression

in separate papers [26, 27]. Second, an interesting extension is to robust models such as the median regression or more generally the quantile regression. Without loss of generality, consider inference on a median regression, we have $\sum_{i=1}^n I(Y_i - \mathbf{X}_i\beta < 0) - \sum_{i=1}^n U_i = 0$, where $U_i \sim \text{Bernoulli}(0.5)$. And a natural choice of the nuclear mapping is $T(\mathbf{U}, \theta) = \sum_{i=1}^n U_i$ if joint inference for β is our target. We refer the readers to the discussions in [59] for more examples and details on the nuclear mapping and repro samples approach. Another research direction concerns weak signals. Although we do not impose any conditions on the signal strength, the proposed approach may demand a high computational cost to recover weak signals. Therefore a natural question is, under limited computational resources, how to adjust the proposed approach for weak signals. Additionally, the identifiability condition or $C_{\min} > 0$ ensures that there is no perfect co-linearity between the true model and an alternative model of equal size. When there is, then multiple equivalent “true” models exist. Our procedure is still valid to cover one of these “true” models. However, constructing a confident set to cover all of these equivalent “true” models is an open problem for future research.

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SUPPLEMENTARY MATERIAL

Supplement to “Finite- and Large-Sample Inference for Model and Coefficients in High-dimensional Linear Regression with Repro Samples”

(). In the Supplementary Materials (Appendices A–F), we provide the theoretical proofs, technical lemmas, additional discussions, and numerical results.

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